

NEURAL BELLMAN OPERATORS: TECHNICAL AND PRESERVATION SUPPLEMENT TO REVISION R9

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This supplement preserves the complete R8 manuscript and proofs, the complete R4 exposition and derivations, and all six delivered R5 numerical tables without altering their source text. It also specifies the R9 replication and evidence map. Recursive utility, temporal selves, dynamic games, Merton benchmarks, stochastic-trace identities, finite safeguards, manufactured-payoff tests, and unsuccessful numerical attempts remain available for review. The new original-economy welfare bounds, constructive certificates, and frozen external comparison are developed in the R9 main paper and its mathematical appendix.

PART I: READING AND REPLICATION MAP

The current manuscript is `ECTA_R9.tex`; its Appendix proves the new results. The accompanying response addresses the September 22, 2026 R8 numerical-methods report at commit `9749c3c`. The new work is isolated on the R9 constructive-certification branch. Earlier main and review branches have not been advanced by this revision.

The R9 replication directory contains the original-payoff actor and moment evaluator, fitted-upper construction and Bernstein verifier, restricted market-deflator verifier, scalar global action certifier, interval neural-jet evaluator, absolute-value relaxation, and frozen external-comparison driver. Numerical tables are generated from their retained JSON records. Every new trained external network and all 4,096-path samples for each of the 72 holdout pairs are retained. Source and preservation hashes, exact commands, scope checks, and failed-pilot status are in the R9 replication and validation records.

The following material is historical but preserved in full. References to a preceding revision, its new experiment, its current target, or its conclusions refer to that historical record. In particular, the R8 manufactured-payoff result is not evidence for original-payoff precision, and its six-seed external comparison is superseded for current inference by the frozen R9 holdout. Neither passage is deleted or silently rewritten. Historical finite safeguards remain distinct from the new continuous-action oracle. The R4 recursive, temporal-self, and game derivations retain their own assumptions and equilibrium concepts; no new large-scale extension result is inferred from the additive-control computations.

1. INTRODUCTION

How accurately must an economic decision rule be computed before a welfare comparison is informative? This question is especially consequential when the decision maker changes not only consumption and investment but also a preference state. A small error in a fitted value function need not imply a small policy loss, and a numerically stable policy surface need not distinguish an adjustment-cost effect from an approximation error. The difficulty is compounded when an economic boundary is an actual termination event rather than an artificial edge of a training domain.

This paper develops Neural Bellman Operators (NBO) as a model-based framework for separating policy evaluation, feasible policy improvement, and verification of the implemented decision rule. The economic application combines stochastic risk aversion, consumption, portfolio choice, and costly preference adjustment. Its stopping and settlement contract is specified as part of the economy. The mathematical results identify three distinct objects: an exact policy-improvement sequence, its neural implementation, and an independently evaluated policy. None is substituted for another in the error bounds.

Our first result provides a continuous-time verification pair. A supersolution controls the value of every admissible policy, whereas a subsolution controls the payoff of the particular policy to be implemented. Their difference, together with explicitly signed residual and boundary terms, bounds regret. The argument uses smooth test functions, not smoothness of the optimal value. It therefore does not require continuity of the optimal value at an irregular controlled exit boundary. We give both a direct continuous-generator implementation and an operator-enclosure decomposition for a discretized implementation. The latter separates time, action, state interpolation, quadrature, exit, inner-solve, and arithmetic errors; a refinement difference is not one of these bounds.

The second result connects computational accuracy to an economic comparative static. Raising the price of preference adjustment lowers the optimal discounted adjustment budget. For policies that are only approximately optimal, the possible reversal in this ordering is bounded by the sum of their regret bounds divided by the change in price. Value enclosures also bound optimal adjustment budgets through secants and bound the welfare gain from access to the adjustment technology. These statements make the required numerical precision depend on the economic comparison, rather than on a common uncalibrated stopping tolerance.

The computational analysis has complementary components. We re-evaluate the retained neural NDU policies, price the full finite-action safeguard, and separate time, state, and action refinements. We construct outward-interval verification pairs for their deployment in the original continuous diffusion. Those pairs are valid but too conservative to resolve the stated economic precision target. A second, explicitly modified-payoff experiment trains a bounded actor on exactly the same state, control, covariance, and exit geometry and obtains an all-state continuous regret certificate. Finally, a coupled nonconvex control problem in eight and sixteen dimensions is compared at a fixed CPU budget with the pinned author implementation of SOC-MartNet. The comparison uses actual neural controls, no analytical greedy-action labels, paired independent policy rollouts, and a declared six-seed design. Its outcome measures realized cost at that budget, not distance to an unknown optimum.

Policy iteration, neural PDE evaluation, and their combination have important antecedents. Jacka and Mijatović (2015) study continuous-time policy improvement. Kim, Cho, Kim, and Kim (2026) alternate soft evaluation and improvement for entropy-regularized stochastic control and separate iteration, policy-network, and residual errors. The updated interior analysis of Kim, Kim, Kim, and Cho (2026) treats stationary second-order equations with bounded-domain

training and unresolved outer information; its interior bounds and its discussion of collocation–closed-loop mismatch are directly relevant. Lee and Kim (2025) combine policy iteration with operator learning to reuse a representation across terminal data. Cai, Fang, and Zhou (2025) train stochastic control and value networks through martingale conditions without explicitly computing the Hamiltonian infimum. NBO does not claim priority for neural policy iteration or for residual verification. Its contribution here is the combination of stopped economic-model verification, explicitly costed feasible correction, and error-calibrated preference-adjustment comparisons. Section 7 gives the comparison at the level of assumptions and implemented experiments.

Recursive utility, sophisticated temporal selves, and dynamic games remain part of the framework. They do not share an identical evaluation equation or an identical optimality notion. We retain their aggregators, equilibrium conventions, proofs, and computational records, and state precisely how a gain bound changes when the evaluation operator is nonlinear or when rivals are frozen. The supplementary material preserves the complete preceding R4 manuscript and technical derivations; the current manuscript integrates the R5 evidence and the new continuous verification and external-comparison results.

2. AN ECONOMY WITH COSTLY PREFERENCE ADJUSTMENT

The state is $s = (u, x)$, where u is relative risk aversion and x is financial wealth. An admissible control is a progressively measurable process $a = (c, \theta, p)$ taking values in

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

Before termination the state satisfies

$$du_t = \theta_t dt + .05 dZ_t^u, \quad (1)$$

$$dx_t = [(.02 + .06p_t)x_t - c_t]dt + .2p_t x_t dZ_t^x, \quad d\langle Z^u, Z^x \rangle_t = -.25 dt. \quad (2)$$

The horizon is $T = 1$, the open state domain is $D = (1.2, 2.8) \times (.5, 2)$, and the discount rate is $\rho = .04$. The process is stopped at $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$. Its running payoff and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad k > 0, \quad (3)$$

$$G(u, x) = -.02(u - 2)^2 + .1 \log x, \quad g(t, u, x) = G(u, x) - 8(1 - t). \quad (4)$$

Thus

$$J_k^\pi(t, s) = \mathbb{E}_{t,s}^\pi \left[\int_t^\tau e^{-\rho(r-t)} \ell_k(S_r, a_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V_k(t, s) = \sup_\pi J_k^\pi(t, s). \quad (5)$$

The feasible controls, state dynamics, stopping rule, and settlement do not depend on k . Bounded controls and locally Lipschitz state coefficients give a well-defined stopped state equation for the feedbacks used below. Bounded payoffs on the stated domain give finite values. The cardinal normalization and consumption units are held fixed across comparisons. A transformation of utility depending on the controlled state u is generally a different economy.

The early termination fee is a specified liquidation covenant, not an estimated preference parameter. At termination there is no subsequent wealth injection or continuation consumption. This distinction is essential: a constant preference diffusion cannot be confined to the stated

rectangle by merely clipping simulated states. At the lower wealth boundary the largest admissible drift is $-.016$, so that the original consumption floor cannot support a silently viable self-financing rectangle. We do not replace this economy by reflection.

At the upper wealth boundary, the admissible choice $p = 0$ has zero wealth diffusion and inward drift $.04 - c < 0$. Consequently, a theorem that assumes boundary regularity uniformly over all controls cannot simply be applied to this model. We neither identify the interior trace of V_k with the prescribed immediate-settlement value without proof nor assume a globally smooth optimal value. The verification pair in Section 4 uses inequalities on the actual stopped trace and remains meaningful in this setting.

For a smooth test function v , define

$$\mathcal{L}^a v = \theta v_u + [(.02 + .06p)x - c]v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux}, \quad (6)$$

$$R_a v = v_t + \mathcal{L}^a v + \ell_k - \rho v. \quad (7)$$

The interior HJB equation is $\sup_{a \in A} R_a V_k = 0$ in the appropriate solution sense. All covariance terms, including the mixed derivative, are retained. A negative shock correlation by itself does not establish the sign of the optimal portfolio response, and dividing a re-solved V_u by k does not establish a pointwise ordering of adjustment policies.

PROPOSITION 1—Complete control selection for a fixed jet: *Let v have the derivatives in (7). A global maximizer of $R_a v$ is obtained separately in the three control components. If $v_x > 0$, $c^* = \text{proj}_{[.05, .8]}(v_x^{-1/u})$; otherwise $c^* = .8$. Also $\theta^* = \text{proj}_{[-.2, .2]}(v_u/k)$. Write*

$$h_1 = .06xv_x - .0025xv_{ux}, \quad h_2 = .04x^2v_{xx}.$$

If $h_2 < 0$, then $p^ = \text{proj}_{[-.5, .8]}(-h_1/h_2)$. If $h_2 \geq 0$, choose an endpoint maximizing $h_1p + h_2p^2/2$, with a fixed tie rule.*

This selection rule includes nonpositive marginal wealth value, zero curvature, convex portfolio objectives, and constrained optima. It removes action discretization from a continuous-generator audit for this particular model. It is not a closed-form optimizer of the killed-Euler, interpolated, finite-step Bellman target, and it is not a global convergence theorem for shared neural parameters. The underlying derivative rule was present in the retained derivations; the present implementation tests every curvature branch explicitly.

3. EVALUATION, IMPROVEMENT, AND FINITE CORRECTION

For a fixed policy π , policy evaluation solves $R_\pi v = 0$ with the economic stopping contract. Improvement chooses a feasible control maximizing the Hamiltonian of that fixed evaluation. The exact operators are denoted $\mathcal{E}$ and $\mathcal{I}$, and exact policy iteration is $\pi_{j+1} = \mathcal{I}(\mathcal{E}\pi_j)$. If a classical evaluation exists, $R_{\pi_{j+1}}(\mathcal{E}\pi_j) \geq 0$. Verification then gives $J^{\pi_{j+1}} \geq J^{\pi_j}$.

An exact limiting argument requires more than this monotonicity. Under compactness of the policy class, continuity of evaluation into a precompact $C^{1,2}$ jet class, closure under improvement, continuity of the chosen selector at cluster points, and comparison for the optimality equation, each cluster-point value is optimal. These are hypotheses on the exact operators. They do not follow from successful stochastic-gradient training. Appendix B gives the limiting argument and retains the distinctions between parameter stationarity, policy convergence, and value convergence.

3.1. The raw neural algorithm

A critic v_ξ is trained against fixed-policy targets. In a continuous-generator implementation these are $R_\pi v_\xi$; in a discrete implementation they are the appropriate one-step conditional expectations. The actor improves a frozen critic. Freezing the critic's parameters does not mean deleting the derivative of its output with respect to the controlled next state. Conversely, no actor update may change the critic parameters or a rival's policy parameters through the improvement objective. Independent policy evaluation is a third operation and supplies neither optimal-action nor optimal-value training labels.

In the NDU implementation inherited from R5, the raw actor has three bounded outputs and two tanh hidden layers. It is trained without enumerating actions, but its training states lie on a lattice. It is therefore a continuous-action, lattice-trained implementation, not a mesh-free experiment. The high-dimensional experiment below instead uses sampled states and Gaussian successor samples. The two implementations share the separation of evaluation and improvement; their verification costs are different.

3.2. A certificate for the declared finite economy

Let $\mathcal{T}_n^a w = r_n^a + \delta P_n^a w$, where P_n^a is a positive substochastic continuation matrix and $\delta = e^{-\rho\Delta}$. Termination payoffs are included in r_n^a . Let $\hat{\pi}$ be a feasible policy and v_n a critic, with terminal discrepancy at most b . Define

$$e_n = \|v_n - \mathcal{T}_n^{\hat{\pi}} v_{n+1}\|_\infty, \quad \eta_n = \|\max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\hat{\pi}} v_{n+1}\|_\infty.$$

THEOREM 1—Finite gain accounting: *On this finite economy,*

$$\|V_{h,n}^* - V_{h,n}^{\hat{\pi}}\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

If $v = V_h^{\hat{\pi}}$ is independently evaluated with the exact finite terminal data, $e_j = b = 0$. A state-dependent bound is obtained by setting $B_N = 0$ and

$$B_n(s) = g_n(s) + \delta \max_a (P_n^a B_{n+1})(s), \quad g_n = \max_a \mathcal{T}_n^a V_{h,n+1}^{\hat{\pi}} - V_{h,n}^{\hat{\pi}}.$$

Then $0 \leq V_{h,n}^ - V_{h,n}^{\hat{\pi}} \leq B_n$.*

Here the maximum must range over the entire declared candidate economy. For the retained NDU audit this is the 125-point action grid together with the frozen continuous neural proposal at every state and date. The certificate does not cover all continuous controls merely because one candidate originated in a neural network. Floating-point computation of the finite envelope is recorded separately from the real-arithmetic statement of the theorem.

3.3. The safeguarded algorithm and its cost

The safeguarded algorithm first evaluates the raw policy, computes all candidate gains, and corrects states with gain above a specified threshold. It re-evaluates the corrected policy and stops only when its reported envelope meets the target. In the new re-audit the threshold is

$$\gamma = \frac{\varepsilon}{2 \sum_{n=0}^{N-1} \delta^n}. \quad (9)$$

The factor two leaves a margin relative to the sufficient condition $\max_{n,s} g_n(s) \leq \gamma$. The actual stopping test remains the recomputed envelope, not an optimal reference value. A stopped run that does not meet that test is a failed precision test.

PROPOSITION 2—When corrections can become infrequent: *For M audited state-date decisions with nonnegative gains g_m , the share whose gain exceeds $\gamma > 0$ is at most $(M\gamma)^{-1} \sum_m g_m$. Thus a vanishing mean gain is sufficient for a vanishing correction share at a fixed threshold. For thresholds $\gamma_j \downarrow 0$, the sufficient condition is $M_j^{-1} \sum_m g_{j,m} = o(\gamma_j)$.*

This is a conditional mechanism, not evidence that longer training has already made the retained corrections negligible. It distinguishes a useful limit statement from an unmeasured training-dose claim. Every audit still enumerates the candidate actions. With N dates, S lattice states, and A grid actions, the implementation uses $NS(A+2)$ value-action evaluations per audit, including policy evaluation and the extra proposal, and $NS(A+1)$ envelope-action evaluations. Each NDU action uses four shock branches and bilinear interpolation. There is no dimension-free cost claim for this safeguard.

4. VERIFICATION OF THE CONTINUOUS ECONOMY

The object implemented in continuous time must be stated before it can be certified. For the retained NDU proposals, we use their feasible bilinear interpolation on the exact uniform state grid. On each decision interval the feedback function is frozen in time, but is evaluated at the current state continuously: $a_r = \pi_n(S_r)$. This is not a simulation that holds the initial action until the next date. It defines a bounded, piecewise-time, Lipschitz state-feedback rule for the original Brownian dynamics and exact first-exit contract.

Write $C_\rho(h) = (1 - e^{-\rho h})/\rho$, with $C_0(h) = h$.

ASSUMPTION 1—Admissible verification witnesses: *U and L are bounded, continuous up to the stopped trace, and $C^{1,2}$ in the interior of each time slab. Their derivatives permit localized Itô verification and the stopped expectations are uniformly integrable. The policy π has a well-defined stopped state equation. At interior slab interfaces either the witnesses are continuous or they satisfy $U(t^-, s) \geq U(t^+, s)$ and $L(t^-, s) \leq L(t^+, s)$.*

THEOREM 2—A one-sided continuous verification pair: *Under Assumption 1, suppose throughout the open domain*

$$\sup_{a \in A} R_a U \leq \epsilon_U, \quad R_\pi L \geq -\epsilon_L,$$

and on the actual terminal and exit trace $U \geq g - b_U$ and $L \leq g + b_L$, where all four constants are nonnegative. Then

$$V(t, s) \leq U(t, s) + b_U + C_\rho(T - t)\epsilon_U, \quad (10)$$

$$J^\pi(t, s) \geq L(t, s) - b_L - C_\rho(T - t)\epsilon_L, \quad (11)$$

$$0 \leq V - J^\pi \leq U - L + b_U + b_L + C_\rho(T - t)(\epsilon_U + \epsilon_L). \quad (12)$$

The true value need not belong to $C^{1,2}$ or be continuous at every controlled boundary point.

The two witnesses need not be neural networks, and need not have the same approximation error. A trained network can propose either witness, but verification requires an enclosure

over the entire stated domain and the full control set. Passing a random test set is not such an enclosure. The symmetric residual certificate is recovered by taking $U = L = v$, with $|R_\pi v| \leq e$, $\sup_a R_a v - R_\pi v \leq \eta$, and $|v - g| \leq b$: regret is at most $2b + C_\rho(2e + \eta)$.

Spatially localized errors can be handled without replacing them by a sampled mean. If $\sup_a R_a U \leq q_U(s, t)$, choose a nonnegative potential z_U with zero-or-positive terminal/exit trace such that $z_{U,t} + \mathcal{L}^a z_U - \rho z_U \leq -q_U$ for every action. Then $U + z_U$ is an exact upper witness. The analogous potential for the fixed policy converts L into a lower witness. These occupation-sensitive potentials identify a constructive route to tighten conservative global certificates; they themselves must be verified rather than estimated on favorable trajectories.

4.1. An explicit original-payoff construction

For G in (4), Proposition 1 gives $c = .8$, $p = .8$, and $\theta = -.02(u - 2)$ when $k = 2$. A 160-by-160 outward-interval partition bounds $\sup_{s,a} R_a G$ above by $-\alpha$, with $\alpha > 0$. For each frozen policy we likewise bound $R_\pi g$ below by $-\beta$ on every state cell and time slab. Consequently,

$$U = G - \alpha C_\rho(1 - t), \quad L = g - \beta C_\rho(1 - t) \quad (13)$$

are valid exact-sign witnesses. Since $\alpha < 8$, $U \geq g$ on the stopped trace, while $L \leq g$. Their date-zero difference is

$$8 + (\beta - \alpha)C_\rho(1). \quad (14)$$

The interval program encloses logarithms, powers, products, and all bilinear-action component ranges. Input network outputs are frozen binary64 constants interpreted as exact real numbers; the state grid is the exact uniform rational grid. The returned bound is rounded outward again when converted to binary64. This certifies the specified real-arithmetic feedback, not every floating-point implementation of the stochastic differential equation.

TABLE I
ORIGINAL-PAYOFF CONTINUOUS NDU WITNESS BOUNDS

N	Interval cells	α	β	Regret upper bound
6	4320	0.864832	7.078878	14.091405
12	8640	0.864832	1.211230	8.339561

Retained raw seed-10 nodal policies, deployed as continuously reevaluated bilinear feedback on each time slab. Bounds cover the original continuous Brownian model and exact first exit using outward interval arithmetic. Both bounds exceed the .01 precision target.

The original-payoff bounds in Table I are numerically wide. They replace an absent continuous certificate by an executed, valid certificate, but do not establish regret below .01 or .05. In particular they are not a substitute for a sufficiently narrow welfare enclosure. The theorem and its implementation make that distinction testable rather than hiding it in a missing discretization term.

4.2. A complete discretization accounting route

There is a second route to continuous verification. Let $\mathcal{B}_n^*$ be the exact continuous-time optimal Bellman semigroup over one decision interval, allowing all admissible within-interval controls, and let $\mathcal{B}_n^\pi$ be the exact semigroup for the declared continuously implemented policy.

Lift the finite optimal and finite policy values to bounded functions w_n^* and w_n^π by a positive norm-one extension. Suppose certified enclosures give

$$\|w_n^* - \mathcal{B}_n^* w_{n+1}^*\|_\infty \leq \kappa_n^*, \quad \|w_n^\pi - \mathcal{B}_n^\pi w_{n+1}^\pi\|_\infty \leq \kappa_n^\pi,$$

with terminal discrepancies b^* and b^π . A telescoping semigroup argument yields

$$\|V_0 - J_0^\pi\|_\infty \leq B_h + \delta^N(b^* + b^\pi) + \sum_{n=0}^{N-1} \delta^n(\kappa_n^* + \kappa_n^\pi), \quad (15)$$

where B_h bounds the lifted finite optimal–policy gap. Each κ can be enclosed by a declared chain of intermediate operators separating time freezing (including within-step feedback), restriction of actions, state interpolation, shock quadrature, first exit and settlement timing, inner optimization, and roundoff. The sum is valid when each adjacent operator difference is bounded on the same lifted continuation function. A null term is unknown, not zero. A coarse–fine solution difference alone bounds none of the uncomputed tails. Nor does a regular-domain scheme theorem automatically apply at an irregular controlled exit boundary. The direct generator pair (12) bypasses these scheme errors by auditing the continuous implemented policy itself; it does not assert that the errors of the training scheme vanish.

5. ECONOMIC COMPARISONS AT FINITE NUMERICAL ACCURACY

Fix an initial state and suppress it from the notation. For each feasible policy write

$$A(\pi) = \mathbb{E} \left[\int_t^\tau e^{-\rho(r-t)} \frac{c_r^{1-u_r}}{1-u_r} dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad B(\pi) = \mathbb{E} \int_t^\tau e^{-\rho(r-t)} \frac{\theta_r^2}{2} dr.$$

Then $V(k) = \sup_\pi \{A(\pi) - kB(\pi)\}$, and $0 \leq B(\pi) \leq .02C_\rho(T-t)$.

THEOREM 3—Adjustment costs and approximate optimality: *If the feasible policies and termination contract are independent of k , V is convex and nonincreasing. For $k_1 < k_2$, optimal policies satisfy $B(\pi_2) \leq B(\pi_1)$. More generally, if $V(k_i) - J_{k_i}^{\pi_i} \leq \varepsilon_i$, then*

$$B(\pi_2) - B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}. \quad (16)$$

If measured budgets satisfy $|\hat{B}_i - B(\pi_i)| \leq \xi_i$, their possible reversal is bounded by the right side of (16) plus $\xi_1 + \xi_2$. At a differentiability point with an attained optimizer, $V'(k) = -B(\pi_k)$.

The coefficient in (16) is sharp without further restrictions on the feasible policy set. The conclusion concerns discounted adjustment expenditure, not pointwise values of $\theta(t, u, x)$. The exact monotonicity statement is an economic theorem for the original continuous contract; the numerical experiment must still establish sufficient accuracy to quantify its magnitude.

PROPOSITION 3—Value intervals, budgets, and access to adjustment: *Suppose $L(k) \leq V(k) \leq U(k)$ and $k_- < k < k_+$. Every attained optimal adjustment budget at k belongs to*

$$\left[\max \left\{ 0, \frac{L(k) - U(k_+)}{k_+ - k} \right\}, \min \left\{ .02C_\rho(T-t), \frac{U(k_-) - L(k)}{k - k_-} \right\} \right]. \quad (17)$$

If $[L_F, U_F]$ and $[L_R, U_R]$ enclose the flexible and the restricted $\theta \equiv 0$ values, the value of access to adjustment belongs to $[L_F - U_R, U_F - L_R]$, intersected with $[0, \infty)$.

These bounds determine meaningful numerical targets. To resolve a budget difference of size d_B across prices separated by Δk , an error budget of order $\Delta k d_B$ is required, together with a budget-evaluation enclosure. An arbitrary .05 value target need not suffice. Likewise, a positive measured flexible-minus-restricted value is not a certified welfare gain unless its lower interval endpoint is positive. The utility units here are explicitly normalized lifetime utility; they have not been converted into an empirically calibrated consumption-equivalent welfare metric.

6. NUMERICAL EVIDENCE

The experiments distinguish numerical validity, achieved precision, and comparative performance. The original-payoff NDU certificate in Table I is an outward-interval bound for a continuous diffusion. The finite gain envelopes in this section are float64 calculations for their declared candidate economies. The learned-geometry experiment has a rigorous continuous verification pair but a deliberately modified running payoff. The external comparison estimates realized costs and sampling uncertainty; it does not know the optimal value. These distinctions are part of the interpretation of the results, not alternative names for the same error.

The complete new panel was executed from source commit 14ecbd48 in a clean Python 3.11 environment, with PyTorch 2.10.0, NumPy 2.3.5, SciPy 1.17.0, and mpmath 1.3.0. Numerical work used one CPU thread. The ledger retains the processor description, exact versions, source hashes, external-code hashes, checkpoint weights, rollout arrays, all seed-level records, and execution logs. Table generation reads those committed records and fails on missing cells. The local integration runs are development checks, not extra independent confirmatory seeds.

6.1. *Preserving a learned policy through thresholded correction*

Table II re-audits two retained policies without retraining or selecting a new policy using the reference optimum. The twelve-date seed-10 policy had a raw localized envelope of .0922420. The preceding all-positive-gain safeguard replaced 54.6602 percent of interior state-date decisions and obtained a .00544535 envelope. The threshold in (9), at target .05, replaces only .8746 percent and obtains a .00682279 envelope. At target .01, the correction share is 4.9350 percent and the envelope is .00676060. Thus most of the earlier replacements were unnecessary for meeting the declared finite-economy tolerance in this sample.

TABLE II
THRESHOLDED CORRECTION OF RETAINED NDU POLICIES

N	Target	Raw bound	Final bound	Override (%)	Audits	Seconds
6	0.05	0.042805	0.042805	0.000	1	0.472
6	0.01	0.042805	0.003278	8.796	2	0.849
12	0.05	0.092242	0.006823	0.875	2	1.641
12	0.01	0.092242	0.006761	4.935	2	1.630

Float64 finite-candidate envelopes, not continuous-model certificates. Seconds include all listed audits and envelope calculations on the current worker. Each audit still enumerates the full 125-action grid plus the proposal. Historical training time is not added across different environments.

This improvement addresses the role of correction rather than concealing it. The corrected rule differs from the raw rule, and every audit still enumerates 125 grid actions plus the neural proposal. For the twelve-date runs, two audits use 2,362,200 value-action evaluations and 2,343,600 envelope-action evaluations, each with four transition branches. The new policy is not

obtained by a cheap constant-cost acceptance test. Table II includes evaluation, gain calculation, envelope propagation, and re-audit time. Historical training time is retained separately because it was measured in a different run and environment; adding it to current audit time would not be a same-machine end-to-end timing experiment.

The six-date policy already passes the .05 target without correction. At .01, it requires an 8.7956 percent correction share. These are deterministic post-review re-audits of the specified seed-10 policies, not a new population estimate of raw training success. Proposition 2 explains the condition under which correction shares vanish, but this two-policy exercise does not demonstrate a training-dose convergence curve. Neither its finite envelope nor its smaller correction share closes the continuous NDU error budget.

6.2. Separating time, state, and action approximation

Table III varies one approximation factor at a time. The time experiment fixes a 25×31 lattice and five points per action component; the state experiment fixes 48 dates and five action points; the action experiment fixes 48 dates and the 25×31 lattice. All retain the original four-sign shock rule and the straight-segment liquidation treatment. The latter two approximations are not independently refined in this panel.

TABLE III
ONE-FACTOR NDU APPROXIMATION DIAGNOSTICS

Factor	N	State nodes	n_a	Value	Exit prob.	h^2/Δ
time	24	25×31	5	-1.451486	0.104315	0.1067
time	48	25×31	5	-1.549787	0.193358	0.2133
time	96	25×31	5	-1.608530	0.227492	0.4267
state	48	25×31	5	-1.549787	0.193358	0.2133
state	48	37×46	5	-1.444048	0.151095	0.0948
state	48	49×61	5	-1.391646	0.102389	0.0533
action	48	25×31	3	-1.606541	0.245260	0.2133
action	48	25×31	5	-1.549787	0.193358	0.2133
action	48	25×31	9	-1.536711	0.182784	0.2133

All entries use $k = 2$ and evaluate $(u, x) = (2, 1.25)$. n_a is the number of nodes per action component. Four-sign quadrature and straight-segment liquidation are unchanged; these differences do not bound continuous-model error.

The time-only sequence is not evidence of convergence. Its central value moves from -1.45149 at 24 dates to -1.60853 at 96 dates, while the exit probability rises substantially. Keeping the spatial mesh fixed makes h^2/Δ increase, rather than shrink; the experiment exposes this interaction instead of fitting an unjustified time order. Conversely, refining the state lattice at 48 dates moves the value from -1.54979 to -1.39165 . Action refinement improves the finite optimum, but the three approximation factors cannot be conflated. The operator-agreement regression is close to machine precision; agreement of two implementations of the same approximation is not an enclosure of its continuous-model bias.

A further panel uses 96 dates, a 49×61 lattice, and nine points in each action component, or 729 actions. Each adjustment-cost economy is re-solved. Table IV shows the same ordering of adjustment expenditure as Theorem 3, with an active preference margin. This is a stronger finite-action resolution than the earlier five-point panel, but still not a verified continuous welfare difference. The first-exit and quadrature terms in (15) remain unbounded by these calculations. Accordingly, neither a Richardson remainder nor a continuous policy-convergence rate is inferred from this table.

TABLE IV
FINER FINITE-ACTION PREFERENCE-ADJUSTMENT COMPARISON

k	Value	Budget B	Exit prob.	θ_0	p_0
0.5	-1.416240	0.016811	0.162725	0.2000	0.3125
2	-1.437379	0.011583	0.162286	0.2000	0.3125
8	-1.468043	0.001888	0.158876	0.1000	0.3125

$N = 96$, a 49×61 state lattice, and nine nodes per action component (729 actions). Central consumption is .8 in these rows. Values and budgets belong to the declared numerical liquidation economy, without a continuous error enclosure.

6.3. A learned actor on the full NDU control and exit geometry

To test whether continuous verification transfers from an analytic example to an actually trained control, retain (2), the three-dimensional action box, correlated shocks, state domain, and settlement exactly. Change only the running payoff to

$$\tilde{\ell}(t, s, a) = \ell_2(s, a) - \sup_{b \in A} R_b g(t, s). \quad (18)$$

The subtracted term is independent of the chosen action, but depends on state and time. Thus it generally changes optimal behavior relative to the original-payoff economy; it is not an innocuous preference normalization. In the modified problem $\sup_a \tilde{R}_a g = 0$ and the explicit feasible maximizer is $(.8, -.02(u - 2), .8)$. Itô verification proves that the modified value is g with the original stopping contract.

We train a four-parameter actor with constant bounded sigmoid consumption and portfolio outputs and an affine-tanh adjustment output. The actor maximizes the Hamiltonian of the known witness, with no optimal-action labels. Six declared seeds each receive 4,000 Adam steps at learning rate .02 and 256 random states per step. This is a controlled verification experiment, not a deep-network capacity or high-dimensional scaling claim. The witness is known rather than learned; the actor and all of its achieved errors are learned.

For the frozen weights, the control gap is bounded over the entire state domain by outward interval arithmetic. The consumption gain is bounded using

$$\int_c^{.8} z^{-u} dz - .1(.8 - c)/x \leq (.8 - c)\{c^{-2.8} - .05\},$$

the portfolio gain is the exact quadratic difference, and the adjustment gain equals $(\theta + .02(u - 2))^2$ at $k = 2$. A 2,048-cell interval partition in u encloses the latter. The full regret is bounded by $C_\rho(1)$ times the summed action-gap enclosure. The time discretization of a rollout and a numerical first-exit probability do not enter this certificate, because its proof is for the original continuous generator.

Every seed meets the .01 modified-problem target, with continuous regret bounds between .00081053 and .00112263. Table V separately reports componentwise control errors rather than treating value accuracy as policy sup-norm accuracy. This is the desired transfer test with a learned actor and the original first-exit/control geometry. It does not identify the original NDU value, and does not tighten Table I merely by sharing the state equation.

TABLE V
ALL-STATE CONTINUOUS CERTIFICATE FOR TRAINED NDU-GEOMETRY ACTORS

Seed	Regret bound	c error	θ error	p error	Train (s)
300	0.00081494	0.0004555	0.0000318	0.0007034	2.774
301	0.00103566	0.0005788	0.0000344	0.0007415	2.749
302	0.00081053	0.0004532	0.0000308	0.0005803	2.778
303	0.00112262	0.0006277	0.0000397	0.0005607	2.761
304	0.00112212	0.0006273	0.0000635	0.0006387	2.794
305	0.00099308	0.0005551	0.0000334	0.0007077	2.780

All-state upper bounds use outward interval arithmetic and the same state, action, covariance, and first-exit geometry as the economic model. Only the running payoff is manufactured. Four actor parameters are trained for 4,000 steps without optimal-action labels; the critic witness is known.

6.4. Coupled nonconvex control and an external author implementation

The new comparison does not use the Cole–Hopf quadratic-control model or regress to an analytic greedy target. Starting from $X_0 = 0$ in $\mathbb{R}^d$, minimize

$$\mathbb{E} \left[\int_0^1 f(a_t) dt + q(X_1) \right], \quad dX_t = 2a_t dt + \sqrt{2}(.4)dW_t, \quad a_t \in [-1, 1]^d, \quad (19)$$

$$f(a) = d^{-1} \sum_{i=1}^d (a_i^2 + .1 \sin^2 a_i) + .3 \sin \left(\sum_i a_i \right), \quad (20)$$

$$q(x) = d^{-1} \sum_i (x_i - .5)^2 + .2 \cos \left(d^{-1/2} \sum_i x_i \right). \quad (21)$$

The sine term couples all control components. For example, at an interior vector whose sum is $\pi/2$, the Hessian of the running cost has a negative direction along the all-ones vector for both tested dimensions. Hence the problem is genuinely nonconvex in the action. No closed-form maximizer or optimal-control labels are supplied to either method, and the quadratic exponential transformation does not eliminate this Hamiltonian.

The external comparator is the unmodified `SOCMartNet.train` routine in the authors’ repository `sx-fang/MartNet`, pinned at commit `991ea8dd`. Its complete module hashes are in the execution manifest. The integration consists of model and network adapters and a scheduler that stops after a completed update at the time budget. The authors’ solver is not rewritten as a bespoke PINN comparator. The pinned implementation uses a Hessian-free Hamiltonian interface; we therefore use constant diffusion in this comparison. This is a restriction of the implemented interface used here, not an assertion that the published `SOC-MartNet` framework excludes volatility control.

Both methods receive identical initial actor and critic weights within each seed. Both use two tanh hidden layers of width 48, bounded actor outputs, a terminal-exact critic, 16 training time intervals, and one CPU thread. NBO uses three evaluation updates per improvement update, 256 sampled states, four Gaussian successors, and Adam with learning rate `.003`. `SOC-MartNet` retains its additional test network, RMSprop updates, and the declared martingale and Hamiltonian losses. Its actor/value learning rate is `.003/ $\sqrt{d}$` , the test-network learning rate is `.01`, and the remaining adapter settings are fixed in the protocol. The test network has 600 outputs; its parameters and all setup costs are reported rather than excluded from the method description.

The primary budget is ten seconds of training per method, allowing the final update to complete. Setup time is outside this primary budget and is reported separately. This matters: the mean setup-plus-training time is approximately 10.69 seconds for NBO and 10.02 seconds for SOC-MartNet on this worker. Therefore the experiment is not described as an exactly matched ten-second end-to-end race. NBO is evaluated first in each process, so one-time framework initialization is included in its measured setup cost. Table VII makes the distinction visible.

TABLE VI
PAIRED NONCONVEX-CONTROL COMPARISON AT A FIXED TRAINING BUDGET

d	NBO cost	SOC cost	Difference	Paired 95% interval
8	0.59567	2.31352	-1.71785	$[-2.13823, -1.29747]$
16	0.48067	2.72799	-2.24732	$[-2.51288, -1.98176]$

Difference is NBO minus SOC-MartNet realized cost at 128 audit steps; negative favors NBO. Each dimension has six independent training/audit seeds, 4,096 common-random-number paths per pair, and ten seconds of training per method. Student intervals describe across-seed uncertainty, not optimality loss. Maximum paired path standard errors are 0.01098, 0.00900, respectively.

TABLE VII
MEASURED TRAINING AND SETUP RESOURCES

d	Method	Train (s)	Setup + train (s)	Outer updates
8	NBO	10.003	10.692	1536.7
8	SOC-MartNet	10.012	10.015	465.3
16	NBO	10.004	10.689	1477.3
16	SOC-MartNet	10.012	10.014	458.0

Single-thread CPU means over six seeds. Identical actor/critic parameter counts within each dimension: 3,224/2,881 at $d = 8$, and 4,000/3,265 at $d = 16$. SOC-MartNet also trains its 600-output test network. Paired policy audit means are approximately .379 and .440 seconds. Outer-update definitions differ across algorithms and are not equal units of work.

At dimension eight, the paired mean NBO-minus-SOC cost is -1.71785 , with a six-seed Student interval $[-2.13823, -1.29747]$. At dimension sixteen it is -2.24732 , with interval $[-2.51288, -1.98176]$. Thus the new evidence favors NBO under the specified adapter and short training budget. This is a demonstrated benefit in a coupled nonconvex problem, in contrast to the retained R5 quadratic-control comparison. It is not a theorem of computational dominance, a replication of the published SOC-MartNet hardware or learning curves, or a statement that either computed policy is near optimal.

6.5. Uncertainty, mesh diagnostics, and retained failures

Each paired policy audit uses 4,096 paths with common Brownian increments for the two policies and an independent seed for each training seed. The primary outcome uses 128 time steps. A nested 64-step audit aggregates adjacent increments, so its difference from the finer audit is a matched time-mesh diagnostic. Table VI reports the maximum paired path standard error separately from the across-seed interval. The interval is based on six independent training/audit seeds, with a Student approximation rather than a distribution-free coverage claim. Paths are not counted as independent training runs.

The average 128-minus-64 cost changes are about $-.00149$ and $-.00243$ for NBO and SOC-MartNet at dimension eight, and $-.00176$ and $.00951$ at dimension sixteen. These changes are much smaller than the observed paired cost differences, but they are not certified bounds on the remaining time-discretization error. A matched-accuracy computational frontier and longer-budget learning curves would answer additional questions not identified by the present fixed-budget design.

All prior outcomes remain part of the evidence. The R5 ten-seed baseline had ten raw finite-target passes, whereas its three twelve-date raw policies failed. Its recursive raw policies missed their own target, and its eight/sixteen/thirty-two-dimensional quadratic-control NBO means were worse than those of the in-house PINN-PI comparator. Those results are neither relabeled as successes nor removed after observing the new comparison. The R4 Merton, nonlinear NDU, cost, boundary, recursive, temporal-self, game, and coupled linear-quadratic calculations, including failed pilots and incomplete runs, are reproduced in the preservation supplement. The new external experiment changes the tested model and comparator; it does not retrospectively change those historical results.

7. RELATION TO NEIGHBORING NUMERICAL METHODS

Table VIII separates the relevant methodological axes. Its entries summarize the cited analyses and the implemented comparison; they are not a ranking by paper title. In particular, an interior PDE error theorem with unknown outer data, a stopped-control regret theorem with a contractual boundary, and a finite candidate-action gain envelope address different mathematical objects.

TABLE VIII
RELATED METHODS AND THE OBJECTS COMPARED

Method	Principal mechanism or guarantee	Relation to the present evidence
Exploratory PINN policy iteration	Soft policy updates for entropy-regularized HJB equations; iteration, policy-network, and PDE residual errors	Neighboring policy-evaluation/improvement analysis; not the external implementation tested here
Interior physics-informed policy iteration	Stationary interior error control with residual and outer-information contributions under stated regularity conditions	Clarifies what interior collocation can control without supplying the stopped NDU boundary data
PI-DeepONet	Policy iteration and reusable operator approximation across terminal data	Representation reuse is a different performance target from one trained policy at a fixed budget
SOC-MartNet	Martingale training of value and control without an explicit Hamiltonian infimum	Pinned author routine tested on coupled constrained nonconvex controls in $d = 8, 16$
NBO in this paper	Separated evaluation, improvement, and independent gain or witness verification	Continuous stopped-model theorem, explicit economic error calibration, finite correction costs, and paired author-code experiment

Sources: Kim, Cho, Kim, and Kim (2026), Kim, Kim, Kim, and Cho (2026), Lee and Kim (2025), Cai, Fang, and Zhou (2025). The new external benchmark uses constant diffusion and a ten-second training budget. Its scope does not establish a general ranking for recursive utilities, games, or higher dimensions.

For an analyst choosing a method, the distinction between state dimension and action dimension is crucial. Neural state representations avoid a tensor state grid during the high-dimensional

training experiment, while the finite NDU safeguard still has tensor action cost. The exact component selection in Proposition 1 is specific to that economic Hamiltonian. The coupled sine-control problem deliberately removes that separability. Likewise, none of the numerical comparisons establishes that a small sampled training loss satisfies the uniform hypotheses of the continuous certificate. The contribution is a verifiable separation of these issues, with one favorable external comparison and one original-economy precision test whose conservative outcome remains visible.

8. RECURSIVE UTILITY AND STRATEGIC DECISIONS

8.1. Recursive utility

Let $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $bV > 0$. The retained Epstein–Zin normalization is

$$f(c, V) = \frac{\rho}{a} c^a (bV)^{1-a/b} - \frac{\rho b}{a} V, \quad (22)$$

with the appropriate logarithmic limit when $a = 0$; see Epstein and Zin (1989), Duffie and Epstein (1992). Evaluation is $V_t + \mathcal{L}^{c,p} V + f(c, V) = 0$, not an additive-reward equation with the preference parameters renamed. The transform $V = \exp(W)/b$ enforces the sign domain. A compact positive margin for bV , bounds on consumption, and a monotone implicit step are additionally needed for a finite Lipschitz bound on the evaluation operator.

The gain argument extends beyond affine evaluation. If the action operators $\mathcal{T}_n^a$ are monotone and uniformly Lipschitz with constants L_n on a verified continuation-value interval, then (8) holds with δ^{j-n} replaced by $\prod_{m=n}^{j-1} L_m$, and the terminal factor by $\prod_{m=n}^{N-1} L_m$. Appendix D proves this statement. A sign-enforcing architecture alone does not verify the required compact interval or monotonicity.

The supplementary record retains the stationary homothetic anchors $p = .3$ and $c/x = .0455$ at the declared parameters, the executed implicit recursive reference, and the R5 trained recursive runs. In the latter, raw losses around .0425 and .0502 against a .01 target were corrected using a 961-action reference. Those corrected results establish finite-model performance of the corrected rule; they do not establish small correction cost or a scaled neural recursive-portfolio solution. These experiments and their limitations are preserved rather than replaced by an additive-control result.

8.2. Sophisticated temporal selves

For a sophisticated acting self with present-bias parameter β and ordinary discount δ , the improvement and evaluation equations differ:

$$c_n(s) \in \arg \max_c \{u(c)\Delta + \beta\delta\mathbb{E}[V_{n+1}(F(s, c, \varepsilon))]\}, \quad (23)$$

$$V_n(s) = u(c_n(s))\Delta + \delta\mathbb{E}[V_{n+1}(F(s, c_n(s), \varepsilon))]. \quad (24)$$

Only improvement carries the extra β . With two consumption dates, terminal log wealth, zero interest, and $\delta = 1$, the sophisticated first-date share is $1/(1 + 2\beta)$, hence .4166667 at $\beta = .7$. Replacing ordinary evaluation by geometric- β evaluation gives .4566210, a different behavioral model. The retained derivation, executed residual check, and feasible-deviation calculation remain part of the supplement. Equilibrium verification concerns each acting self's feasible deviation with future selves fixed, not a fictitious commitment planner who can change all future selves' preferences.

8.3. Dynamic games

For player i , fix the rivals' feedbacks and solve both the induced policy-evaluation problem and a unilateral best-response problem over all subsequent dates and states. Applying the finite or continuous gain certificate to this best-response problem bounds player i 's exploitability. A Nash certificate requires this test for every player. Differentiating joint profits, allowing gradients through rivals, or checking only realized-path deviations is a different operation.

The retained static Cournot and three-date stochastic capacity-game records test these distinctions. The dynamic game has two firms, capacity investment, stochastic demand, and multiple equilibrium selections. The R5 neural game adds actual actor training, but its eighteen states and analytic best-response verification remain small-scale. Neither experiment establishes general equilibrium scalability or a competitive limit. The operator formulation, nonzero present bias, recursive aggregator, and strategic state transitions are retained as substantive extensions; larger certified experiments on these objects remain a separate quantitative requirement, not a consequence of the new additive nonconvex benchmark.

9. COMPUTATION WITH STOCHASTIC TRACES

If A is symmetric and z_k are independent standard Gaussian probes, put $\bar{q}_K = K^{-1} \sum_k z_k' A z_k$. Then

$$\mathbb{E}(c - \bar{q}_K/2)^2 = (c - \text{tr } A/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (25)$$

Averaging separately squared one-probe residuals retains a variance penalty that does not decrease with K . Two independent probe means give an unbiased cross-product estimator of the squared exact residual, but not a nonnegative sample objective or a uniform residual enclosure. The proof, variance accounting, and the retained Monte Carlo regression are given in the supplement and Appendix E.

For a diffusion loading with m columns, an exact contraction can use m directional second derivatives; a randomized contraction uses K probe operations and the loading operations. Network size, parameter differentiation, collocation coverage, optimization iterations, and the accuracy-dependent probe budget all enter total cost. Avoiding a dense Hessian is useful, but does not imply a universal dimensional crossover or eliminate the cost of verification. The retained coupled linear-quadratic tests with off-diagonal drift, rewards, and shocks are useful regression checks, while their known quadratic value family and faster Riccati comparator preclude a general performance conclusion.

10. CONCLUSION

NBO makes a policy-evaluation approximation, an improvement rule, and a verification calculation separately inspectable. In a stopped preference-portfolio economy, this separation leads to a complete constrained Hamiltonian audit, one-sided continuous value witnesses, and quantitative conditions for adjustment-cost comparisons at finite numerical accuracy. The new learned-geometry certificate and paired author-code comparison address different questions: transfer of a verification method and realized performance on coupled nonconvex controls. Neither is used to erase the conservative original-payoff certificate or the substantial cost of a finite safeguard.

The economic scope continues to include recursive utility, sophisticated temporal selves, and strategic interaction, with their distinct evaluation and equilibrium conditions. The remaining numerical task is concrete: construct sufficiently narrow original-model witnesses, or enclose

every term of the declared semigroup error budget, so that the economic effect exceeds the verified approximation error. Complete current source, retained derivations, executed records, and a source-bound build make this requirement available for the next independent review.

APPENDIX A: PROOFS OF THE CONTINUOUS VERIFICATION RESULTS

A.1. Proof of Theorem 2

For a smooth witness w and a fixed admissible policy a , localized discounted Itô calculus gives

$$\mathbb{E} \left[e^{-\rho(\sigma-t)} w(\sigma, S_\sigma) + \int_t^\sigma e^{-\rho(r-t)} \ell(S_r, a_r) dr \right] = w(t, s) + \mathbb{E} \int_t^\sigma e^{-\rho(r-t)} R_{a_r} w(r, S_r) dr, \quad (26)$$

where σ first stops on a compact interior localization or at the economic stopping time. Localization also bounds the stochastic integral. The hypotheses on the witness and uniform integrability permit passage to τ ; no regularity of the optimal value is invoked.

For $w = U$, $R_a U \leq \epsilon_U$ for every admissible action. The terminal inequality $g \leq U + b_U$ and $e^{-\rho(\tau-t)} \leq 1$ yield

$$J^a(t, s) \leq U(t, s) + b_U + \epsilon_U \mathbb{E} \int_t^\tau e^{-\rho(r-t)} dr \leq U(t, s) + b_U + C_\rho(T-t)\epsilon_U.$$

Take the supremum over policies; attainment is unnecessary. For $w = L$, use the specified policy π , the inequalities $R_\pi L \geq -\epsilon_L$ and $g \geq L - b_L$, and the same integration identity to obtain the lower bound. Subtraction proves the regret bound, and nonnegativity follows from the definition of the supremum.

For multiple time slabs, apply (26) on each slab and sum. Forward-time jumps contribute $U(t^+, S_t) - U(t^-, S_t)$ to the upper-witness identity. They are nonpositive under Assumption 1, so the upper inequality is preserved. Lower-witness jumps have the opposite sign. This explains why a piecewise-time approximation may not silently omit its interface errors. The actual policy can be only piecewise continuous in time because Itô's formula is applied to the witnesses, not to a derivative of the policy. *Q.E.D.*

A.2. Spatial residual potentials

Suppose the upper residual satisfies $R_a U \leq q_U$ and a nonnegative potential obeys

$$z_{U,t} + \mathcal{L}^a z_U - \rho z_U \leq -q_U \quad \text{for every } a.$$

Then $R_a(U + z_U) \leq 0$ by the affine dependence of R_a on the witness. Its boundary inequality is preserved by nonnegativity of z_U . Similarly, if $R_\pi L \geq -q_L$ and $z_{L,t} + \mathcal{L}^\pi z_L - \rho z_L \leq -q_L$, then $R_\pi(L - z_L) \geq 0$. Theorem 2 applies to $U + z_U$ and $L - z_L$. A potential solving an occupation problem exactly would weight residuals by reachable discounted occupation; an approximate potential is usable only after the displayed differential inequality is enclosed. This construction does not turn an empirical occupancy histogram into a theorem.

A.3. Verification of the explicit original-economy witnesses

Since $\partial_t C_\rho(1-t) = -e^{-\rho(1-t)}$ and $\rho C_\rho(1-t) = 1 - e^{-\rho(1-t)}$,

$$(\partial_t - \rho)[-a C_\rho(1-t)] = a$$

for a constant a . Hence $\sup_a R_a G \leq -\alpha$ implies $\sup_a R_a U \leq 0$, and $R_\pi g \geq -\beta$ implies $R_\pi L \geq 0$. On the stopped boundary,

$$U - g = 8(1 - t) - \alpha C_\rho(1 - t) \geq (8 - \alpha)(1 - t) \geq 0, \quad L - g = -\beta C_\rho(1 - t) \leq 0.$$

Both witnesses equal G at maturity. They are smooth and bounded on the closed state rectangle because $x \geq .5$. The bilinear policy is globally Lipschitz on this compact rectangle, component-wise feasible, and piecewise constant in its time index. The stopped coefficients are bounded and have the required state regularity, so the discounted stochastic integrals are integrable. The sign enclosure therefore proves a statement about the original continuously stopped policy, including policies for which the optimal boundary regularity conditions of a different PDE theorem fail.

The interval calculation bounds all a in the upper witness through the exact maximizing rule for G . It bounds the lower witness on every time-state cell by the convex range of the four policy corners, enlarged to a Cartesian box; enlarging the action range can only make the lower bound more conservative. Interval dependency can likewise enlarge the bound, never certify a narrower true range. No assertion of floating-point simulation conformance is needed: the mathematical policy is defined by the stored constants and exact interpolation weights. A separate deployment-error allowance would be required for another implementation.

A.4. Proof of Proposition 1

The action-dependent part of $R_a v$ is the sum of

$$\frac{c^{1-u}}{1-u} - v_x c, \quad v_u \theta - \frac{k}{2} \theta^2, \quad h_1 p + \frac{h_2}{2} p^2.$$

The mixed state derivative changes h_1 but introduces no product between control components. The consumption objective has derivative $c^{-u} - v_x$ and strictly negative second derivative $-u c^{-u-1}$. If $v_x \leq 0$, its derivative is positive everywhere. Otherwise its unique unconstrained stationary point is $v_x^{-1/u}$, and projection gives the constrained optimum. The adjustment objective is a strictly concave quadratic because $k > 0$. The portfolio objective is strictly concave when $h_2 < 0$, linear when $h_2 = 0$, and convex when $h_2 > 0$; the stated vertex or endpoint rule is complete in all cases. Compact product feasibility and additivity make these separate maximizers a global joint maximizer. Q.E.D.

A.5. A learned-geometry regret bound

For (18), let $H_g^*(t, s) = \sup_a R_a g(t, s)$ for the original running payoff. The modified residual is $\tilde{R}_a g = R_a g - H_g^* \leq 0$, with equality for the feasible explicit optimizer. Applying (26) first to any policy and then to this optimizer proves $\tilde{V} = g$. For the learned policy,

$$g(t, s) - \tilde{J}^\pi(t, s) = \mathbb{E}_{t,s}^\pi \int_t^\tau e^{-\rho(r-t)} \{H_g^* - R_\pi g\}(r, S_r) dr.$$

The action-independent terms cancel. At $k = 2$ the adjustment term completes the square to $(\theta + .02(u - 2))^2$. The other two action differences are exactly those bounded in Section 6.3. Their all-state interval upper bound times $C_\rho(1 - t)$ gives the recorded continuous regret certificate. Policy sup-norm errors are separate interval consequences, not inferred by reversing a regret inequality. Q.E.D.

APPENDIX B: POLICY ITERATION AND FINITE GAIN ACCOUNTING

B.1. *The exact separated-operator limit*

Let $V_j = \mathcal{E}\pi_j$ and $\pi_{j+1} = \mathcal{I}(V_j)$. The selector inequality implies $R_{\pi_{j+1}}V_j \geq 0$, so exact comparison and verification yield $V_{j+1} \geq V_j$. Precompactness of the evaluation image bounds the sequence. Its pointwise increasing limit is denoted $\bar{V}$.

Every subsequence has a further subsequence converging in the declared norm of the function, time derivative, gradient, and Hessian. The function component of any such limit must equal $\bar{V}$, by pointwise monotonicity. Its derivative components are the derivatives of that same function limit. Thus the whole value sequence converges in that strong norm: otherwise a subsequence separated from the unique possible strong cluster point would contradict precompactness.

Along a policy subsequence $\pi_{j_m} \rightarrow p$, continuity of evaluation gives $\mathcal{E}p = \bar{V}$. Continuity of the selector at this cluster point gives $\pi_{j_m+1} \rightarrow \mathcal{I}(\bar{V})$, and continuity of evaluation gives $\mathcal{E}(\mathcal{I}(\bar{V})) = \bar{V}$. The evaluation equation of the improved limiting policy therefore reads

$$R_{\mathcal{I}(\bar{V})}\bar{V} = 0.$$

The defining maximizing property of that selector implies $\sup_a R_a \bar{V} = 0$. Boundary comparison identifies $\bar{V} = V^*$. The proof establishes equality of limiting values, not equality of consecutive limiting policies. Compactness, selector closure, and strong evaluation continuity are hypotheses, not properties of an arbitrary neural architecture or Adam trajectory. *Q.E.D.*

B.2. *Proof of Theorem 1*

Positivity and substochasticity give $\|\mathcal{T}_n^a v - \mathcal{T}_n^a w\|_\infty \leq \delta \|v - w\|_\infty$. The maximized operator has the same Lipschitz constant. Since the candidate set includes the enacted action,

$$\|v_n - \max_a \mathcal{T}_n^a v_{n+1}\|_\infty \leq e_n + \eta_n.$$

Comparing this approximate equation with optimal backward induction yields

$$\|v_n - V_{h,n}^*\|_\infty \leq \delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} (e_j + \eta_j).$$

Comparing instead with policy evaluation gives the same inequality with only e_j in the sum. Add the two bounds to prove (8).

For the localized bound let $d_n = V_{h,n}^* - V_{h,n}^\pi$. Exact evaluation implies $g_n \geq 0$. For every state, the elementary maximum-difference inequality gives

$$d_n \leq g_n + \delta \max_a P_n^a d_{n+1}.$$

Starting with $d_N = 0$, induction and positivity give $d_n \leq B_n$. The reference optimal value is unnecessary in this recursion: only policy evaluation, transition matrices, and feasible gains are used. *Q.E.D.*

B.3. *Threshold correction and arithmetic scope*

If every recomputed gain is at most γ , Theorem 1 gives a regret bound at most $\gamma \sum_n \delta^n$. Thus (9) is sufficient with margin when this uniform condition holds. Correcting a subset of decisions does not by itself guarantee that condition, since the re-evaluated continuation changes. The

implementation therefore re-evaluates and re-audits rather than declaring success after a fixed number of replacements. Its stopping certificate is the actual envelope of the corrected policy.

For Proposition 2, let I_m indicate $g_m > \gamma$. Then $\gamma I_m \leq g_m$, so $M^{-1} \sum_m I_m \leq (M\gamma)^{-1} \sum_m g_m$. The fixed- and shrinking-threshold implications follow directly. This statement measures the correction share under the audit measure; it does not imply that all reachable-state losses have small occupation-weighted averages or that a full audit is computationally free. *Q.E.D.*

The finite envelopes reported in the numerical tables are float64 quantities with independent operator-agreement and regression tests. They are not outward-rounded enclosures of every finite arithmetic operation. To make a machine-rigorous finite certificate one must add roundoff and inner-solve enclosures to e_n and η_n , or evaluate their analogues with validated arithmetic. The continuous witness and learned-geometry certificates do use outward interval arithmetic, and are identified separately for this reason.

B.4. Proof of the continuous semigroup error decomposition

With settlement fixed inside each step, the exact surviving-continuation map has Lipschitz constant at most δ : survival probabilities are at most one and continuation is received at the full step end. Taking a supremum over all admissible within-step controls preserves this bound. Write $E_n^* = \|V_n - w_n^*\|_\infty$ and $E_n^\pi = \|J_n^\pi - w_n^\pi\|_\infty$. Then

$$E_n^* \leq \kappa_n^* + \delta E_{n+1}^*, \quad E_n^\pi \leq \kappa_n^\pi + \delta E_{n+1}^\pi.$$

Backward iteration and the positive norm-one lift of the finite gap prove (15). If $\mathcal{B}^{(0)}, \dots, \mathcal{B}^{(m)}$ form an intermediate operator chain from the exact semigroup to the lifted numerical operator, the triangle inequality bounds its discrepancy on a specified continuation w by $\sum_i \|(\mathcal{B}^{(i)} - \mathcal{B}^{(i+1)})w\|_\infty$. Every term must have a verified bound; changing the tested continuation between adjacent differences without controlling that change invalidates the telescoping argument. *Q.E.D.*

APPENDIX C: PROOFS FOR ECONOMIC COMPARISONS

For a fixed policy, $A(\pi) - kB(\pi)$ is affine and nonincreasing in k . The supremum is therefore convex and nonincreasing. The budget is bounded by $.02C_\rho(T - t)$ because $|\theta| \leq .2$. If π attains the value at k , for every other price k' ,

$$V(k') \geq A(\pi) - k'B(\pi) = V(k) - (k' - k)B(\pi).$$

Thus $-B(\pi)$ is a subgradient of V at k . At a differentiability point every attained optimizer has budget $-V'(k)$.

C.1. Proof of Theorem 3

Approximate optimality and feasibility of the rival candidate policy give

$$\begin{aligned} A(\pi_1) - k_1 B(\pi_1) &\geq A(\pi_2) - k_1 B(\pi_2) - \varepsilon_1, \\ A(\pi_2) - k_2 B(\pi_2) &\geq A(\pi_1) - k_2 B(\pi_1) - \varepsilon_2. \end{aligned}$$

Adding and collecting terms yields $(k_2 - k_1)\{B(\pi_2) - B(\pi_1)\} \leq \varepsilon_1 + \varepsilon_2$. The exact monotonicity statement is the zero-error case. Adding the two budget measurement discrepancies yields the stated observable-error inequality. Feasible-set independence of k is essential: otherwise a policy used in the crossed comparison might not be available.

For sharpness in the class of affine policy-value problems, take two policies with $B_1 = 0$, $B_2 = (\varepsilon_1 + \varepsilon_2)/(k_2 - k_1)$, $A_1 = 0$, and $A_2 = k_1 B_2 + \varepsilon_1$. At k_1 , policy 1 is exactly ε_1 suboptimal; at k_2 , policy 2 is exactly ε_2 suboptimal. Equality holds in the budget-reversal bound. Choose errors small enough to respect any prescribed positive upper bound on B_2 . This construction proves that the coefficient cannot be improved without more structure; it is not a claim that these two artificial policies have been realized in the particular NDU dynamics. *Q.E.D.*

C.2. Proof of Proposition 3

Use the subgradient inequality with the optimizer at k and prices k_+ and k_- . Rearrangement gives

$$\frac{V(k) - V(k_+)}{k_+ - k} \leq B(\pi_k) \leq \frac{V(k_-) - V(k)}{k - k_-}.$$

Replacing each numerator by its appropriate outer value enclosure gives (17), and intersecting with the primitive budget range is valid. Inclusion of restricted policies in the flexible set gives $V_F - V_R \geq 0$. The interval subtraction rule gives the other endpoints. No optimizer differentiability, common selected policy, or pointwise action ordering is required. *Q.E.D.*

APPENDIX D: RECURSIVE AND EQUILIBRIUM GAIN BOUNDS

Suppose a verified interval of continuation values is invariant under all action operators, each operator is monotone on it, and each has Lipschitz constant at most $L_n \geq 0$. The maximized operator is also L_n -Lipschitz. Repeating the two approximate-equation comparisons in the finite proof gives

$$\begin{aligned} \|V_n^* - v_n\|_\infty &\leq e_n + \eta_n + L_n \|V_{n+1}^* - v_{n+1}\|_\infty, \\ \|V_n^\pi - v_n\|_\infty &\leq e_n + L_n \|V_{n+1}^\pi - v_{n+1}\|_\infty. \end{aligned}$$

Iteration proves the product-weighted error formula stated in the text. These hypotheses must be verified for the actual recursive discretization. In particular, the sign condition $bV > 0$ does not exclude arbitrarily large derivatives of the driver close to zero and does not supply a uniform L_n .

For a game, fix rivals' strategies and regard player i 's unilateral deviations over all future dates as a control problem with its own payoff and transition law. The preceding certificate then bounds $V_i^{\text{BR}(\pi_{-i})} - V_i^\pi$. Requiring this bound for every player gives an approximate Nash condition at every initial state included in the norm. Checking only a single-date action deviation or only states visited in one rollout does not replace the all-future-date best-response calculation.

For sophisticated temporal selves the acting self's criterion uses the β -weighted continuation in (24), but ordinary continuation evaluation uses δ . A unilateral deviation certificate must use the acting self's criterion, with future selves held fixed. Substituting a geometrically β -discounted evaluation equation changes that equilibrium problem, so its small residual would certify the wrong object. The preserved supplement supplies the full two-date algebra and the executed finite-model conventions. *Q.E.D.*

APPENDIX E: TRACE ESTIMATION AND LIMITATIONS OF LOCAL TRAINING

For a symmetric matrix A and $z \sim N(0, I)$, diagonalize A orthogonally. Rotational invariance and independence of the coordinates imply $\mathbb{E} z' A z = \text{tr } A$ and $\text{Var}(z' A z) = 2 \|A\|_F^2$. The mean

of K independent probes has variance $2\|A\|_F^2/K$. Bias–variance decomposition of $(c - \bar{q}_K/2)^2$ proves (25).

Averaging the K quantities $(c - z'_k A z_k/2)^2$ leaves the expectation equal to $(c - \text{tr } A/2)^2 + \|A\|_F^2/2$. With two independent probe means $\bar{q}$ and $\bar{q}'$, independence instead gives

$$\mathbb{E}[(c - \bar{q}/2)(c - \bar{q}'/2)] = (c - \text{tr } A/2)^2.$$

The latter sample objective may be negative and may have high variance. None of these expectation identities is a uniform residual enclosure over state space. *Q.E.D.*

A local actor stationary point also cannot substitute for a global feasible action maximum. For $H(a) = -(a^2 - 1)^2 + .2a$ on $[-2, 2]$, the local maximum near $-.973994$ and the global maximum near 1.024120 are both isolated stationary points, with objective gap about $.399875$. Exact critic evaluation and isolated actor stationary points do not eliminate this difference. Concavity in the action would give the global supporting-hyperplane gap, but concavity in an action argument does not imply concavity in shared network weights.

The retained viscosity counterexample has the same logical role. A small integrated strong residual for a smooth approximation of $|x| - 1$ does not select the viscosity solution $1 - |x|$ of the appropriate eikonal boundary problem. Missing viscosity inequalities cannot be replaced by a loss plot. The previous derivations, exact boundary specifications, and regression checks for these examples are reproduced unchanged in the preservation supplement.

Likewise, a two-timescale stochastic-approximation theorem requires bounded iterates, appropriate regularity, stable fast evaluation dynamics, martingale-noise control, and step-size conditions. Its generic conclusion is an invariant-set statement for the limiting slow dynamics. An optimal-policy conclusion needs an additional argument ruling out nonoptimal attractors. The current regret theorems do not rely on asserting that every observed gradient trajectory has such an optimal limit.

PART III: RETAINED COMPLETE R4 EXPOSITION

1. INTRODUCTION

Dynamic economic models increasingly place the formation of preferences, the allocation of financial risk, and the adjustment of physical resources inside the same decision problem. The relevant computational object is then a controlled operator: a policy changes both the payoff and the law under which continuation utility is evaluated. A useful approximation must preserve this distinction. In particular, reducing a joint scalar training objective need not improve an economic policy, and a small error in a value representation need not identify the policy at a nearly flat optimum.

This paper develops Neural Bellman Operators (NBO) as a separated evaluation–improvement method for such problems. The critic approximates the continuation value of a specified feasible policy. The actor improves that policy against a frozen continuation object. A third, independent calculation evaluates the enacted policy and its feasible deviation gains. We study both the continuous generator formulation and a positive finite-horizon operator formulation. The latter makes the stochastic transition and the action maximization explicit and provides a reference against which a trained neural representation can be assessed.

There are three substantive results. First, we give a quantitative connection between policy evaluation, boundary approximation, feasible improvement, and loss of economic value. The continuous-time bound applies to a stopped controlled diffusion and requires uniform, rather than merely sampled, residual bounds. Its finite-horizon counterpart is directly computable on a finite state–action approximation. Second, we establish a comparative-static result for endogenous preference formation. When the adjustment cost enters as a price of a discounted adjustment budget, an increase in that price reduces the optimal budget. This conclusion holds for any optimizers and does not require differentiability of the value function or a pointwise sign restriction on preference adjustment. Third, we implement the separated method, including automatic differentiation of a continuous HJB benchmark and trained nonlinear actor and critic networks for a stochastic preference–portfolio problem. Independent policy evaluation separates value accuracy from agreement of the full policy vector.

The economic application preserves the interior preference and wealth dynamics of the original model. It completes the model at the boundary by an explicit early-liquidation contract. The preference coefficient has nonzero diffusion, so a bounded drift cannot keep it in a closed interval with probability one. Likewise, the original consumption floor is incompatible with a self-financing wealth process constrained to the lower endpoint. Liquidation is an economically different boundary condition from reflection or a capital injection; it is specified as such, including the settlement fee and its sensitivity. No transition replenishes wealth and then continues consumption.

The numerical evidence has distinct roles. A continuous-time Merton benchmark tests the HJB derivatives and the separated gradient graph. A two-state NDU problem tests a trained nonlinear neural representation against an independent exhaustive stochastic Bellman solver. A re-solved adjustment-cost panel evaluates the economic comparative static. Recursive utility, sophisticated temporal selves, and a dynamic capacity game exercise additional operator interfaces. A coupled stochastic linear–quadratic problem checks dynamic cross-state dependence at dimensions 4, 8, and 16. These experiments retain the general economic scope without conflating an algebraic identity, a finite-model equilibrium, and a trained solution of a continuous nonlinear economy.

1.1. *Relation to the literature*

The closest methodological antecedents already combine policy improvement and neural approximation. Kim, Kim, Kim, and Cho (2025) study physics-informed neural policy iteration

with policy-error analysis. Lee and Kim (2024) combine Hamilton–Jacobi policy iteration with deep operator learning. Cai, Fang, and Zhou (2025, preprint) learn value and control through a martingale formulation without requiring an explicit Hamiltonian optimizer. Direct neural approximation of stochastic control policies is developed by Han and E (2016). Continuous-time policy improvement and its hypotheses are analyzed by Jacka and Mijatović (2015). Accordingly, neural approximation, operator factorization, and the absence of an analytical maximizer are not claimed here as inventions in themselves.

The contribution relative to these approaches is the particular combination of boundary-aware value-loss control, an economically interpretable preference-adjustment result, and an executable separation of the evaluation, improvement, and policy-certification objects. The finite-operator realization also permits a distinction between errors of neural representation and errors of the numerical economy. The bounds below are not presented as a priority claim for residual-based verification in general; they specify the conditions and constants used by this implementation and its application.

Our economic formulation connects portfolio choice (Merton, 1969), recursive preferences (Epstein and Zin, 1989, Duffie and Epstein, 1992), and the motivating endogenous-preference model of Qi (2025). Grid and projection methods remain important comparators in computational economics (Judd, 1998, Brumm and Scheidegger, 2017). In our two-state experiment, exhaustive backward induction is faster than neural training. That result is reported, not converted into an unobserved scalability advantage. A neural representation can nevertheless be evaluated as a function approximation to a controlled operator, and an independently measured value loss supplies a meaningful test of that representation.

The distinction from Deep BSDE methods (Han, Jentzen, and E, 2018) is about the equations and training objects, not about whether an analyst can write a closed-form control. Semilinearity concerns dependence on the highest-order derivative. Eliminating a control analytically can leave nonlinear Hessian dependence. Conversely, an admissible BSDE representation need not be a Pontryagin adjoint system. Section 7 states the differentiation costs conditionally, including the diffusion rank and the accuracy demanded of a trace approximation.

2. THE ECONOMIC PROBLEM AND ITS BOUNDARY

Let S be a Markov state in an open bounded domain $D \subset \mathbb{R}^d$, let $A(s)$ be a compact feasible action correspondence, and let π be an admissible feedback. Up to $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$, the state satisfies

$$dS_r = b(r, S_r, \pi_r) dr + \Sigma(r, S_r, \pi_r) dW_r. \quad (1)$$

Admissibility includes existence of the stopped process and integrability of the stated payoff. For a time-separable payoff with nonnegative discount rate ρ , define

$$J^\pi(t, s) = \mathbb{E}_{t,s} \left[\int_t^\tau e^{-\rho(r-t)} \ell(r, S_r, \pi_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V^* = \sup_\pi J^\pi. \quad (2)$$

The settlement g is data of the problem, not an output of a projection routine. At T , it equals the terminal endowment utility. The operator is

$$\mathcal{L}^a v = b^a \cdot \nabla v + \frac{1}{2} \text{tr}(\Sigma^a (\Sigma^a)' D^2 v), \quad R_a(v) = v_t + \mathcal{L}^a v + \ell^a - \rho v.$$

Where a classical representation is valid, fixed-policy evaluation has $R_\pi(V^\pi) = 0$, while optimality has $\sup_a R_a(V^*) = 0$. Both have the same declared parabolic-boundary data. A reflected model would instead include regulators and an appropriate derivative boundary condition; the results are not obtained by replacing that condition with a value penalty.

ASSUMPTION 1—Well-defined evaluation and verification: *The feasible controls yield a stopped state process; the payoff and stochastic integrals used below are integrable. For each fixed policy under discussion, evaluation is well defined. Comparison and the relevant verification inequalities hold for the specified generator and boundary operator. Whenever derivatives are used, the test function belongs to $C^{1,2}$ up to the relevant stopped boundary and its discounted Itô formula is valid.*

These hypotheses state the mathematical object to which a residual refers. They do not assert that an arbitrary smooth network or an arbitrary compact computational rectangle verifies them.

2.1. Recursive utility and Markov closure

For recursive preferences, fixed-policy utility is defined before optimization. On a domain of its value argument where the driver is admissible, it solves

$$-dY_r^\pi = f(r, S_r, \pi_r, Y_r^\pi) dr - Z_r^\pi dW_r, \quad Y_\tau^\pi = g(\tau, S_\tau). \quad (3)$$

Existence, uniqueness, comparison, and integrability of this policy-wise equation are prerequisites for taking its supremum. The Markov state must contain any variable on which the driver or future opportunities depend. A general continuous martingale is not replaced by a Gaussian shock merely because its quadratic variation is known; the executed models here use the stated Brownian or finite Markov specifications. The recursive extension and its sign domain are developed in Section 6.1.

3. SEPARATED NEURAL BELLMAN OPERATORS

In the continuous realization, a critic v_ξ is trained for a fixed enacted policy by its evaluation residual and the appropriate boundary loss. The actor parameters ψ are updated against a detached critic jet:

$$L_E(\xi; \pi) = \mathbb{E}_\nu R_\pi(v_\xi)^2 + \lambda_B \mathbb{E}_{\nu_B} |v_\xi - g|^2, \quad (4)$$

$$L_I(\psi; \xi) = -\mathbb{E}_{\nu_I} \mathcal{H}^{\pi_\psi} [\text{stopgrad}(v_\xi, \nabla v_\xi, D^2 v_\xi)]. \quad (5)$$

Feasibility is imposed by the actor parameterization or by explicit constrained maximization. A penalty for a general constraint is an approximation and must be accompanied by a feasibility error. During actor improvement no parameter gradient reaches ξ . In a game no player- i gradient reaches the parameters of a frozen rival.

The actor loss is a local improvement surrogate. It is not generally the lifetime-objective gradient under an arbitrary sampling measure. Such an identification requires the appropriate occupancy or adjoint weighting and differentiability conditions. Likewise, minimizing a weighted sum of (4) and (5) over both parameter blocks solves a different optimization problem. Appendix E retains the composite-loss and viscosity counterexamples and states what the approximation argument does establish.

3.1. A positive finite-operator realization

For the numerical economy, let $\mathcal{S}_h$ be a finite state set and A_h a finite feasible action set. Liquidation proceeds are included in r_n^a and surviving transitions in a substochastic kernel P_n^a . With $\delta = e^{-\rho\Delta}$,

$$T_n^a w = r_n^a + \delta P_n^a w, \quad T_n w = \max_{a \in A_h} T_n^a w. \quad (6)$$

The factor δ applies to surviving full steps; an intra-step liquidation has its own fractional-step discount in r_n^a . Boundary states are settled immediately.

The finite NBO algorithm works backward. It freezes the next critic, computes each feasible action's continuation using the declared stochastic operator, and trains the current actor against these action values. It then trains the current critic to evaluate the actor's enacted hard action. The numerical experiment uses a cross-entropy term for the maximizing action of the *neural continuation*, together with its softmax-weighted action-value regret. Optimal reference values or labels never enter this training loss. At deployment the actor chooses its hard argmax, not a mixture. All continuation targets are detached. This is a finite-horizon neural Bellman factorization, not a claim that one Adam update performs exact policy iteration.

The reference solver is an independently coded NumPy backward induction with exhaustive action evaluation and positive bilinear interpolation. The neural operator is implemented separately in PyTorch. An operator-agreement test precedes training. The trained policy is then re-evaluated by the reference operator; this evaluation is different from reading the critic's prediction. All maxima used for finite-model certificates range over the entire stated finite set.

3.2. Exact improvement and quantitative approximation

PROPOSITION 1—Exact separated-operator limit: *Let the policy space Π be compact in the uniform topology. Let evaluation E be continuous into a precompact subset of $C^{1,2}$, endowed with the uniform norm on the function, its time derivative, gradient, and Hessian. Assume a maximizing selector I maps Π into itself, is continuous at accumulation points, and satisfies the comparison and verification conditions of Assumption 1. If $\pi_{j+1} = I(\pi_j)$, every accumulation point p satisfies $E(p) = V^*$, and the common limiting value solves the HJB boundary problem.*

The decisive step is equality of limiting *values*, not equality of successive limiting policies. The full proof is in Appendix A. Closure under the exact selector is a real hypothesis and is not automatic for a finite neural class.

THEOREM 1—Boundary-aware policy-value certificate: *Under Assumption 1, let v be a test function and π a feasible policy. Suppose, uniformly on the stopped domain,*

$$|R_\pi(v)| \leq \epsilon, \quad 0 \leq \sup_{a \in A(s)} \{R_a(v) - R_\pi(v)\} \leq \eta, \quad |v - g| \leq b \quad \text{on the parabolic boundary.}$$

Then, with $C_\rho(h) = (1 - e^{-\rho h})/\rho$ and $C_0(h) = h$,

$$0 \leq V^*(t, s) - J^\pi(t, s) \leq 2b + C_\rho(T - t)(2\epsilon + \eta). \quad (7)$$

No smoothness of V^ , and no claim that an actor stationary point is a maximizer, is required for this inequality.*

The boundary term is not discounted at the terminal date: exit can occur earlier. Theorem 1 turns three separately meaningful errors into a value loss. A sampled mean-square loss does not supply its uniform hypotheses. A verified residual Lipschitz constant and a covering radius can bridge a finite audit to a uniform bound; such constants must be established, not inferred from a plot.

THEOREM 2—Finite-horizon neural-policy certificate: *For the finite operators (6), let $\hat{v}_N$ have terminal error at most b . For an enacted policy π , define*

$$e_n = \|\hat{v}_n - T_n^\pi \hat{v}_{n+1}\|_\infty, \quad \eta_n = \|T_n \hat{v}_{n+1} - T_n^\pi \hat{v}_{n+1}\|_\infty.$$

Then

$$\|V_n^* - V_n^\pi\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

If the enacted policy is re-evaluated exactly on this finite model, take $\hat{v} = V^\pi$; then $e_j = b = 0$ and the same bound uses only the independently evaluated feasible improvement gaps.

This theorem bounds a finite model, not automatically the underlying diffusion. Time, interpolation, quadrature, action coverage, and boundary-crossing errors remain distinct. An additional verified comparison between the finite and continuous economies would be needed to add them to (8).

3.3. Global action improvement versus parameter stationarity

The Hamiltonian $H(a) = -(a^2 - 1)^2 + .2a$ on $[-2, 2]$ has a stable local ascent attractor near $-.973994$ and a global maximum near 1.024120 . Their payoff gap is $.3998746$. Both stationary points are isolated. Thus an exact fast critic, diminishing two-timescale steps, and isolated actor stationary points do not imply a policy-improvement fixed point.

A usable sufficient condition concerns the *action* maximization. If H is differentiable and concave on a convex compact feasible set, then

$$H^* - H(a_0) \leq \max_{a \in A} \langle \nabla H(a_0), a - a_0 \rangle. \quad (9)$$

The right-hand side is a global supporting-hyperplane gap and handles boundary optima. Point-wise concavity does not imply concavity in shared neural parameters. Our finite implementation therefore audits the maximizing finite action directly. The separate asymptotic stochastic-approximation statement, with its invariant-set conclusion and additional optimality conditions, is given in Appendix E.

4. ENDOGENOUS PREFERENCES AND PORTFOLIO CHOICE

The preference state u is the coefficient of relative risk aversion and X is financial wealth. The controls are consumption c , preference adjustment θ , and the risky portfolio share π . Before liquidation,

$$du = \theta dt + .05 dZ^u, \quad (10)$$

$$dX = [(.02 + .06\pi)X - c]dt + .2\pi X dZ^X, \quad d\langle Z^u, Z^X \rangle = -.25 dt. \quad (11)$$

The domain is $(1.2, 2.8) \times (.5, 2)$ and the control box is $[.05, .8] \times [-.2, .2] \times [-.5, .8]$. The horizon is one year and $\rho = .04$. Utility is cardinally fixed as

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad G(u, X) = -.02(u-2)^2 + .1 \log X.$$

Consumption units and this normalization are held fixed throughout the cost panel. An affine transformation that depends on the controlled preference state generally changes the economic problem.

4.1. The liquidation contract

The process stops when it first reaches the domain boundary or the horizon. Settlement is

$$g(t, u, X) = G(u, X) - F(1 - t), \quad F = 8. \quad (12)$$

The early termination fee represents the utility cost of closing the investment and preference-adjustment arrangement before maturity. This is an explicit covenant/liquidation specification, not a calibrated estimate of such a fee. Its size is varied separately below. The fee vanishes at maturity and does not depend on k . No additional wealth is supplied after settlement because the problem has terminated.

This completes rather than disguises the boundary economics. Constant preference diffusion rules out the earlier viable-rectangle interpretation. At $X = .5$, even the largest wealth drift under the original action box is $-.016$. Keeping the original interior process and the original control floor therefore requires exit, reflection with an explicit regulator, or a different resource model. We choose exit. Economic statements about this contract must not be interpreted as statements about a silently reflected self-financing economy.

The interior HJB is

$$0 = V_t + \sup_{c, \theta, \pi} \left\{ \frac{c^{1-u}}{1-u} - \frac{k}{2} \theta^2 + \theta V_u + [(.02 + .06\pi)X - c]V_X - \rho V \right. \\ \left. + \frac{1}{2}(.05)^2 V_{uu} + \frac{1}{2}(.2\pi X)^2 V_{XX} - .25(.05)(.2)\pi X V_{uX} \right\}, \quad (13)$$

with (12) on the parabolic boundary. All three second-order contributions are retained.

4.2. Policies and economic interpretation

Where the maximizer is interior and the indicated derivatives have the required signs,

$$c = V_X^{-1/u}, \quad \theta = V_u/k, \quad \pi = -\frac{.06V_X}{.2^2 X V_{XX}} - \frac{.05(-.25)V_{uX}}{.2X V_{XX}}. \quad (14)$$

These are candidates, not substitutes for constrained maximization. Consumption and adjustment have concave action objectives. The portfolio objective is concave only when $V_{XX} \leq 0$; otherwise endpoint comparisons are essential. Appendix B gives the full selection rule, including zero curvature and nonpositive marginal wealth value.

The cross derivative produces a hedge against preference shocks correlated with returns. Its sign is endogenous: neither a negative shock correlation nor an assumed positive V_{uX} alone establishes a global portfolio comparative static. Likewise, dividing V_u by k does not determine how the re-solved derivative changes with k . The following result gives a global economic conclusion without either shortcut.

THEOREM 3—The price of preference adjustment: *Suppose the feasible policy set, stopping rule, and settlement contract do not depend on k . Write*

$$A(\pi) = \mathbb{E} \left[\int_t^\tau e^{-\rho(r-t)} \frac{c_r^{1-u_r}}{1-u_r} dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad B(\pi) = \mathbb{E} \int_t^\tau e^{-\rho(r-t)} \frac{\theta_r^2}{2} dr.$$

Then $V(k) = \sup_\pi \{A(\pi) - kB(\pi)\}$ is convex and nonincreasing. If $k_1 < k_2$ and optimal policies π_1, π_2 exist, $B(\pi_2) \leq B(\pi_1)$. At a differentiability point with an attained optimizer, $V'(k) = -B(\pi_k)$. The flexible-preference value weakly exceeds the value under the feasible restriction $\theta \equiv 0$.

The theorem concerns the optimal discounted adjustment budget, not a pointwise ordering of $\theta(t, u, X)$. The same proof applies to the finite stochastic economy, so the numerical panel tests a precise economic restriction rather than an unverified policy-surface narrative.

5. NUMERICAL IMPLEMENTATION AND RESULTS

5.1. Continuous HJB benchmark

We first solve the stationary Merton economy with CRRA coefficient 2, discount .04, riskless return .02, excess return .06, and volatility .2. A homothetic critic is $v(x) = -\exp(w)/x$ and a two-logit actor represents $m = c/x \in [.005, .07]$ and $\pi \in [-.5, 1]$. First and second wealth derivatives are obtained by automatic differentiation. Critic optimization minimizes the scaled fixed-policy HJB residual. Actor optimization holds these derivatives fixed. Neither loss receives the analytical optimal controls or value as a label.

Independent evaluation gives $A^\pi = 1/(mD)$ for $J^\pi = -A^\pi/x$, where $D = .04 + .02 + .06\pi - m - .04\pi^2 > 0$. The optimum has $m^* = .04125$, $\pi^* = .75$. Across seeds 0, 1, and 2 the executed actor recovers m within 8.25×10^{-8} and π within 8.29×10^{-10} . The maximum independently evaluated value loss on $[-.5, 2]$ is 4.70×10^{-9} . The scaled HJB residual is at most 1.38×10^{-8} . A graph assertion verifies that actor backpropagation leaves the critic gradient disconnected. This is a homothetic neural benchmark with three trained parameters, not evidence for a deep high-dimensional approximation theorem. Its value is that the continuous generator, automatic derivatives, evaluation, and actor graph can all be checked independently.

5.2. Stochastic NDU transition and neural architecture

For the preference–portfolio model, four equiprobable branches use independent signs $z_1, z_2 \in \{-1, 1\}$:

$$\Delta u = \theta \Delta + .05\sqrt{\Delta} z_1, \quad (15)$$

$$\Delta X = [(.02 + .06\pi)X - c]\Delta + .2\pi X \sqrt{\Delta} [- .25z_1 + \sqrt{1 - .25^2} z_2]. \quad (16)$$

If the straight segment exits, it settles at its first intersection and terminates; otherwise continuation uses positive bilinear interpolation. The running payoff is integrated with the exact discount factor for the elapsed fractional step while holding the current flow fixed. This defines a killed-Euler numerical economy, not exact Brownian first-passage integration. Interior drift and covariance tests are run before boundary corrections.

The nonlinear actor and critic each have two hidden layers of width 48 with tanh activations. The actor has 125 outputs, representing five values for each of the three controls. The critic uses normalized states and log distances to the four boundary faces; it is represented as $G + (1 - t)N_\xi$ away from boundary nodes, with exact prescribed settlement at boundary nodes. The finite model does not require this boundary-switched representation to be a globally C^2 function. Consequently, the numerical certificate applied to it is Theorem 2, not an unverified use of Theorem 1.

At each of six dates, training uses all interior nodes of a 25×31 lattice, 500 actor Adam steps and 1,000 critic Adam steps, followed by critic L-BFGS refinement. The learning rate is .004. Next-date critics and actor targets are frozen. This experiment is a lattice-trained finite-operator realization, not a mesh-free sampling experiment. Seeds 0, 1, and 2 are reported without selecting the best seed.

TABLE I
INDEPENDENT EVALUATION OF THE TRAINED NDU POLICY

Quantity	Seed 0	Seed 1	Seed 2
Maximum date-zero value loss	.011291	.023526	.012797
Mean interior date-zero value loss	.001583	.001803	.001540
Maximum critic-policy value difference	.144195	.070268	.104457
Finite gain-based upper bound	.259331	.264491	.520503
Training seconds	19.039	19.609	20.737

Values compare the independently re-evaluated hard actor with exhaustive optimization on the identical finite stochastic economy. The gain bound uses maxima at all dates and is conservative. Times are local single-thread CPU measurements, not universal complexity constants.

All three completed runs meet the stated finite-model date-zero value-loss target of .05. The critic's own error is larger than the enacted policy loss, illustrating why a critic prediction is not a policy-evaluation certificate. Boundary discrepancies are at most 4.80×10^{-7} in the float32 neural output. For seed 0 the maximum component differences in (c, θ, π) from the finite optimal action are (.375, .2, .975); the corresponding interior mean differences are (.00183, .00335, .11028). The portfolio policy is therefore not uniformly accurate even when value regret is small. Every seed's complete component vector is retained in the ledger.

Exhaustive reference backward induction takes about .60 seconds in the seed-0 comparison, versus 19.04 seconds for training and .65 seconds for independent policy evaluation. There is no measured speed advantage in this two-state task. The result instead establishes that a genuinely trained actor and critic implement the stated stochastic operator with small independently evaluated finite-model policy loss. Whether representation reuse or higher-dimensional nonlinear models reverse the cost comparison requires matched experiments on those objects.

The initial neural pilot failed: its maximum policy value loss was 4.10, its critic error 4.49, and its boundary representation was inaccurate. The failure and its executed source are retained. Two attempted twelve-date runs reached training output but timed out before completing their final audit; they are recorded as incomplete, not successful runs. These records are not included in the three-seed success count.

5.3. Re-solved preference-adjustment costs

The cost panel uses twelve dates, the 25×31 state lattice, and the 125-action stochastic operator. Each k is independently re-solved, along with the restricted $\theta = 0$ economy. Table II evaluates $(u, X) = (2, 1.25)$. The budget falls by approximately 86 percent between $k = .5$ and

TABLE II
COSTLY PREFERENCE FORMATION UNDER THE LIQUIDATION CONTRACT

	$k = .5$	$k = 2$	$k = 8$
Value	-1.351522	-1.372708	-1.404420
Discounted adjustment budget B	.016520	.011921	.002309
Value of permitting adjustment	.064480	.043293	.011582
Date-zero adjustment θ	.2	.2	.1
Exit probability	.248935	.249938	.249920
Interior share with nonzero θ	.896927	.764993	.475637

The budget includes the factor $\theta^2/2$ and time discounting. Exit probabilities refer to this numerical liquidation contract. Nonzero-action shares average over all decision dates and interior lattice states, not over a population.

$k = 8$. Value monotonicity, budget monotonicity, and dominance over the zero-adjustment policy hold at all audited finite states and dates, up to the implementation tolerance. The preference margin is economically active rather than grid-locked. The result is conditional on the cardinal normalization and settlement contract; it is not identification of these primitives from observed consumption or portfolios.

5.4. Discretization and boundary sensitivity

Separate refinements are deliberately not labeled solution-error certificates. At the same central state, values for (N, n_u, n_x, n_a) are

$(6, 13, 16, 5)$	-1.423383
$(12, 13, 16, 5)$	-1.564151
$(12, 25, 31, 5)$	-1.372708
$(24, 25, 31, 5)$	-1.451486
$(24, 49, 61, 5)$	-1.344458
$(12, 25, 31, 7)$	-1.359628

Here n_a is the number of values per control. The changes are material. Refining time alone while holding a positive interpolation grid fixed need not improve a diffusion approximation: the interpolation contribution to a generator can scale as h^2/Δ . Action refinement and boundary-crossing approximation also matter. We therefore report a verified finite-model neural approximation and a re-solved finite-model cost panel, but not a numerically certified continuous-diffusion value error. The analytical comparison in Theorem 3 does apply to the continuous contract under its stated existence conditions.

Changing the settlement fee from 8 to 6 or 10, with the twelve-date baseline discretization held fixed, changes central value to -1.363909 or -1.379622 . These are different contracts rather than alternative numerical corrections to the same contract. The complete exit and adjustment calculations are retained so that subsequent boundary refinements can be evaluated without treating post-correction feasibility as a zero-error result.

6. RECURSIVE UTILITY AND EQUILIBRIUM EXTENSIONS

6.1. Recursive utility

Let $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $bV > 0$. In the normalization used here, the Epstein–Zin continuous-time aggregator is

$$f(c, V) = \frac{\rho}{a} c^a (bV)^{1-a/b} - \frac{\rho b}{a} V, \quad (17)$$

with the logarithmic limit when $a = 0$. The architecture $V = \exp(W)/b$ enforces the sign domain when $b \neq 0$. A compact positive margin for bV , along with consumption bounds, supplies local driver regularity; a sign assertion without executing the transform is not a domain audit.

The full portfolio evaluation equation is $V_t + \mathcal{L}^{c, \pi} V + f(c, V) = 0$. Formal stationary homothetic ratios at $(\rho, \gamma, \psi, \mu, r, \sigma) = (.04, 5, 1.5, .08, .02, .2)$ are $\pi = .3$ and $m = .0455$. An algebraic ratio is distinct from a verified infinite-horizon solution. We retain these anchors and add an executed finite-horizon implicit recursive-utility calculation. The latter has horizon .5, ten dates, 101 wealth points and 80 consumption fractions, and solves each fixed-action implicit value equation on its negative-value branch. Its maximum implicit-equation residual is 2.46×10^{-13} ; the minimum bV is .0625; the executed terminal sign transform differs from the terminal target

by at most 8.89×10^{-16} . Its continuation outside the wealth grid uses the specified homothetic numerical boundary data. This calculation is a recursive reference, not a trained neural portfolio solution. Appendix C supplies the equation, normalization, and verification requirements for the broader recursive application.

6.2. *Sophisticated temporal selves*

For discrete dates and $\delta = e^{-\rho\Delta}$, a sophisticated acting self takes the continuation policy of future selves as given:

$$c_n^*(s) \in \arg \max_c \{u(c)\Delta + \beta\delta\mathbb{E}[V_{n+1}(F(s, c, \varepsilon))]\}, \quad (18)$$

$$V_n(s) = u(c_n^*(s))\Delta + \delta\mathbb{E}[V_{n+1}(F(s, c_n^*(s), \varepsilon))]. \quad (19)$$

Only improvement carries the extra factor β . For two consumption dates followed by terminal log wealth, zero interest and $\delta = 1$, ordinary continuation has coefficient 2 on log wealth at the first decision. The sophisticated initial consumption share is $1/(1 + 2\beta)$. At $\beta = .7$ this is .4166667, whereas geometric- β evaluation gives .4566210. The corrected executed evaluation residual is below 8.89×10^{-16} and a dense feasible one-shot search finds no positive gain. Strict concavity and the analytical first-order condition establish global optimality for this particular interior log example; the finite search alone would provide only a lower bound on exploitability. The $\beta = 1$ check is retained but is not used to detect an error that vanishes at $\beta = 1$.

6.3. *Dynamic capacity competition*

For player i , a frozen-rival policy defines both the evaluation operator and an independent unilateral best-response problem. The static Cournot benchmark has Nash output $1/3$ and joint-profit output $1/4$; an automatic-differentiation test confirms that the rival gradient is absent from player i 's actor update.

We also solve a genuinely dynamic finite game. Two firms have capacities in $\{0, .5, 1\}$, output in half-unit increments up to capacity, and investment $i \in \{0, 1\}$ that adds .5 capacity when feasible. Demand intercept is $z \in \{2.5, 3.5\}$ with transition matrix having diagonal .8 and off-diagonal .2. Per-period profit is $q_i(z - q_i - q_j) - .2i$, discount is .96, and the horizon is three production dates. Backward enumeration selects a pure Nash equilibrium in every state; lexicographic selection resolves multiplicity in three state-date problems. The maximum feasible one-shot deviation gain is computed as zero in this finite game. The full profile and selection rule are stored, including asymmetric equilibria. This is not a neural dynamic-game training result or a competitive-limit calculation. The latter would require a market-size normalization and a separately solved limiting economy; the operator formulation and its economic questions are retained in the appendix.

7. STOCHASTIC TRACES AND COUPLED DYNAMICS

For symmetric A and Gaussian probes z_k , write $q_k = z_k' A z_k$ and $\bar{q}_K = K^{-1} \sum q_k$. The squared-residual objective satisfies

$$\mathbb{E}(a - \bar{q}_K/2)^2 = (a - \text{tr } A/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (20)$$

Averaging $(a - q_k/2)^2$ instead leaves a variance penalty independent of K . With $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ and $a = 3.5$, the desired expectation is $1 + 7.5/K$, not 8.5 for every K . The replication runs 30,000

independent batches for each $K \in \{1, 2, 8, 64\}$ and records Monte Carlo standard errors. All desired-objective mean checks fall within five reported standard errors. Appendix D gives an unbiased two-probe cross-product correction and explains why unbiased trace estimation does not imply an unchanged training objective after squaring.

A dense Hessian need not be materialized. For diffusion loading with m columns, the exact contraction is a sum of m directional second derivatives; stochastic contraction requires K such probe operations plus loading costs. Network size, parameter differentiation, sample count, optimization iterations, and the probe budget needed for accuracy all enter total cost. There is no model-independent crossover at state dimension 50 or 100.

Finally, the coupled dynamic check has $d \in \{4, 8, 16\}$, eight dates, $x_{n+1} = Ax_n + a_n + S\varepsilon_{n+1}$, $A = .8I + .04 \text{ Adj}$, $Q = I + .1211'/d$, $R = .3I$, and $S = .1I + .0211'/d$. A linear neural actor is improved against a frozen quadratic policy evaluator, and compared with an independent Riccati solution. Off-diagonal transition, payoff, and shock terms are retained. Maximum held-out value errors at the three dimensions are respectively 6.60×10^{-14} , 1.55×10^{-14} , and 3.32×10^{-14} . This is an unconstrained linear-quadratic dynamic regression, not the earlier static quadratic program. Riccati evaluation is faster, and the known quadratic value family makes the test structurally favorable. No constrained sparse-grid superiority or general high-dimensional complexity conclusion follows from it.

8. CONCLUSION

NBO treats policy evaluation, feasible improvement, and independent economic evaluation as different computational objects. The distinction yields explicit value-loss bounds and prevents a critic residual or an actor stationary point from being mistaken for a solution certificate. In the preference-portfolio application, an explicit liquidation contract makes the stochastic model and its numerical transition agree; the re-solved adjustment-cost panel implements a global comparative static for the discounted preference-adjustment budget. Actual neural updates, policy evaluation, diffusion moments, temporal-self recursion, trace estimands, and rival-gradient conventions are now executable.

The economic scope includes recursive utility and strategic interaction without assigning them identical mathematical hypotheses. Further accuracy must be earned through the declared error terms: the current NDU runs certify the finite stochastic economy, while continuous-diffusion refinement remains material. All historical models, derivations, figures, and prior reports remain in the repository, with a retention map distinguishing corrected equations, supplementary material, archival arrays, and executed evidence. This structure permits the next review to assess the mathematics and the numerical economy rather than reconstruct their intended relationship.

APPENDIX A: PROOFS AND THE SCOPE OF THE VALUE CERTIFICATES

A.1. *Proof of Proposition 1*

Write $V_j = E(\pi_j)$. The pointwise selector inequality is $\mathcal{H}^{I(\pi_j)}[V_j] \geq \mathcal{H}^{\pi_j}[V_j]$. Comparison and the assumed verification inequality imply $V_{j+1} \geq V_j$. Precompactness of the evaluation image implies boundedness. Therefore at each state-time point the monotone sequence has a limit $\bar{V}$.

Every subsequence of V_j has a further convergent subsequence in the declared strong norm. Its uniform function limit must be $\bar{V}$, by monotonicity and pointwise uniqueness of the limit. Consequently the whole sequence converges uniformly to $\bar{V}$; in fact it converges in the declared strong norm because otherwise a subsequence separated from its unique possible strong cluster point would contradict precompactness. Derivatives of any strong cluster point are the derivatives

of its function limit. This observation includes the time derivative, which must be controlled when passing an evaluation equation to the limit.

Let $\pi_{j_m} \rightarrow p$ along a policy subsequence. Continuity of evaluation gives $E(p) = \bar{V}$. Continuity of I at p and of E gives

$$E(I(p)) = \lim_m E(I(\pi_{j_m})) = \lim_m V_{j_m+1} = \bar{V}.$$

The policy $I(p)$ attains the Hamiltonian maximum evaluated at $E(p) = \bar{V}$. Its evaluation at this same value therefore gives

$$-\bar{V}_t = \mathcal{H}^{I(p)}[\bar{V}] = \sup_{a \in A} \mathcal{H}^a[\bar{V}].$$

The boundary condition passes to the limit in the declared topology. Comparison identifies the value, and verification gives optimality. In particular $E(p) = V^*$, even if p and $I(p)$ differ on ties. No step identifies successive limiting *policies* merely from a small difference of their values.

This proposition concerns an exact operator on a compact, selector-closed policy class. Neither a finite tanh network nor a finite-sample optimizer automatically meets those assumptions. The implemented finite policy is instead assessed by Theorem 2. For recursive drivers the comparison assumptions have to be supplied for that driver and value domain; the proof does not obtain comparison from the word “recursive.”

A.2. Proof of Theorem 1

Apply discounted Itô’s formula to the test function under any feasible policy a and stop at τ . Taking expectations yields the identity

$$J^a(t, s) - v(t, s) = \mathbb{E}_{t,s} \left[e^{-\rho(\tau-t)} (g - v)(\tau, S_\tau) + \int_t^\tau e^{-\rho(r-t)} R_a(v)(r, S_r) dr \right]. \quad (21)$$

If necessary, localization followed by the integrability hypothesis justifies removal of the localizing sequence. For the enacted policy, $R_\pi(v) \geq -\epsilon$ and the boundary difference is at least $-b$. Hence $J^\pi \geq v - b - C_\rho(T - t)\epsilon$. For every feasible alternative, $R_a(v) \leq R_\pi(v) + \eta \leq \epsilon + \eta$ and the boundary difference is at most b . Thus $J^a \leq v + b + C_\rho(T - t)(\epsilon + \eta)$. Taking a supremum over alternatives and subtracting proves (7). Attainment of that supremum is not required.

The proof does not need differentiability of the true value or differentiability of the policy with respect to its parameters. It does need a legitimate test function, a legitimate stopped process, and bounds valid wherever any competing policy can travel. A residual measured under the actor’s occupancy alone need not control an alternative policy. This is why an off-policy uniform audit, or an explicitly proved change-of-measure bound, is essential for the displayed worst-case guarantee.

A.2.0.1. Boundary and domain qualifications. The theorem bounds the stopped payoff as defined, even when the optimal value lacks a classical extension to every boundary face. The NDU diffusion can be degenerate in wealth under $\pi = 0$. We do not assert that all boundary points are regular for every controlled diffusion, or that the continuous NDU value belongs to $C^{1,2}$ up to all faces. The scalar payoff at a boundary starting state is settlement by definition; an interior limiting value may require a generalized viscosity boundary interpretation. The theorem’s test-function and residual hypotheses would still have to be verified before numerical use. The finite boundary-switched critic is not used to bypass this requirement.

A.2.0.2. *Sampling and covering errors.* Suppose a residual R is known to be L_R -Lipschitz on a compact set and audit nodes form a d_h -net. Then

$$\|R\|_\infty \leq \max_j |R(s_j)| + L_R d_h.$$

The same argument applies to a feasible improvement gap if its Lipschitz constant is established. Finite action coverage gives an additional $L_A h_A$ term only when a uniform action Lipschitz bound L_A is known. These inequalities are deterministic covering statements, not consequences of a large sample size. No such global constants have been computed for the NDU neural critic, and no continuum certificate is reported for it.

A.3. Proof of Theorem 2

Positivity and row sums at most one imply that T_n^π and T_n are monotone and δ -Lipschitz in the sup norm. Let $d_n^\pi = \|\hat{v}_n - V_n^\pi\|_\infty$. The evaluation defect gives $d_n^\pi \leq e_n + \delta d_{n+1}^\pi$, and hence

$$d_n^\pi \leq \delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} e_j.$$

For the optimal value, the residual with respect to T_n is at most $e_n + \eta_n$ in absolute value, because $\hat{v}_n - T_n \hat{v}_{n+1} = (\hat{v}_n - T_n^\pi \hat{v}_{n+1}) - (T_n \hat{v}_{n+1} - T_n^\pi \hat{v}_{n+1})$. The same backward estimate gives

$$\|\hat{v}_n - V_n^*\|_\infty \leq \delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} (e_j + \eta_j).$$

The triangle inequality proves the result. For exact re-evaluation, both e and b vanish; the code computes each date's maximum feasible gain and accumulates it with the factor δ .

Early liquidation creates no additional continuation coefficient: its payoff, including the fractional-step discount, is already in r_n^a . The substochastic surviving row sums remain at most one. Boundary nodes are replaced by their immediate settlement operator in both the optimizer and policy evaluator. The theorem therefore applies to the finite model even though its boundary policy outputs are not economically enacted.

A.3.0.1. *Adding a verified numerical-economy error.* Suppose, for the same policies and initial states, an independent argument establishes $|J^\pi - J_h^\pi| \leq \zeta$ uniformly and $|V^* - V_h^*| \leq \zeta_*$. Then $V^* - J^\pi \leq \zeta_* + (V_h^* - J_h^\pi) + \zeta$. This decomposition identifies the missing bridge without assigning it a zero. In particular, a coarse-fine value difference is neither ζ nor ζ_* without a further error argument.

A.4. Proof of Theorem 3

For each admissible policy, $J_k(\pi) = A(\pi) - kB(\pi)$ is affine and nonincreasing in k , since $B \geq 0$. The supremum of these affine functions is convex; the pointwise supremum of nonincreasing functions is nonincreasing. At an initial state where optimizers exist, their optimality gives

$$\begin{aligned} A(\pi_1) - k_1 B(\pi_1) &\geq A(\pi_2) - k_1 B(\pi_2), \\ A(\pi_2) - k_2 B(\pi_2) &\geq A(\pi_1) - k_2 B(\pi_1). \end{aligned}$$

Adding gives $(k_2 - k_1)[B(\pi_1) - B(\pi_2)] \geq 0$. This holds for any choices of the two optimizers, including multiple optimal policies. Moreover $-B(\pi_k)$ is a subgradient of V at k because $V(\tilde{k}) \geq V(k) - (\tilde{k} - k)B(\pi_k)$. At a differentiability point this subgradient equals $V'(k)$. Finally the unrestricted feasible policy set contains the set with $\theta = 0$, so its supremum is no smaller.

The argument survives the killed finite approximation because its transition, discounting, and settlement do not depend on k , while the per-action cost remains exactly k times its discounted adjustment exposure. It would not apply without modification if changing k also changed the liquidation fee, the feasible action set, the utility normalization, or the law of preference shocks. It says nothing about a pointwise ordering of θ , a portfolio-share ordering, or identification of the model from observed data.

APPENDIX B: GLOBAL ACTION SELECTION AND THE NDU STOCHASTIC OPERATOR

B.1. Constrained Hamiltonian maximization

At a fixed critic jet, the NDU maximand separates across c, θ, π . Let their bounds be (c_-, c_+) , (θ_-, θ_+) , and (π_-, π_+) . Consumption maximizes $c^{1-u}/(1-u) - V_X c$, whose second derivative is $-uc^{-u-1} < 0$. If $V_X > 0$, its optimum is the projection of $V_X^{-1/u}$ onto $[c_-, c_+]$. If $V_X \leq 0$, the objective is increasing and the optimum is c_+ . Adjustment maximizes $V_u \theta - k\theta^2/2$ and therefore selects the projection of V_u/k .

Write the portfolio objective as $a_1\pi + a_2\pi^2/2$, where

$$a_1 = .06XV_X + .2(.05)(-.25)XV_{uX}, \quad a_2 = .2^2X^2V_{XX}.$$

If $a_2 < 0$, select the projected value $-a_1/a_2$. If $a_2 > 0$, compare the two endpoints. If $a_2 = 0$, choose the endpoint indicated by the sign of a_1 , with any feasible action optimal when $a_1 = 0$. Thus a nonconcave portfolio slice need not create an inaccessible local search problem in this particular model. The finite algorithm instead enumerates its declared action grid; it never labels a stationary neural parameter as a global action optimum.

For a general differentiable concave maximand, concavity gives $H(a) \leq H(a_0) + \langle \nabla H(a_0), a - a_0 \rangle$. Taking a supremum proves (9). If an interior maximizer and an m -strongly concave objective are known, the familiar gradient-square bound $H^* - H(a_0) \leq \|\nabla H(a_0)\|^2/(2m)$ follows by completing the square. The supporting-hyperplane formulation is preferable at constrained optima. Neither argument transfers concavity from actions to shared network parameters.

B.2. Local consistency and preference reachability

Let $b = (\theta, (r + e\pi)X - c)$ and let C denote the covariance matrix of the two-dimensional diffusion. Before the exit correction, (16) satisfies

$$\mathbb{E}\Delta S = b\Delta, \quad \text{Cov}(\Delta S) = C\Delta, \quad C_{uX} = \sigma_u\sigma_X\pi X\rho_{uX}.$$

The six polynomial tests $1, u, X, u^2, X^2, uX$ give the correct constant and linear generators exactly, and quadratic-generator errors $O(\Delta)$ from the drift outer product. At $(u, X) = (2, 1)$ and $(c, \theta, \pi) = (.4, .13, .65)$, the continuous generators are

$$(0, .13, -.341, .5225, -.6651, -.553625).$$

The executable checks step sizes $.1, .05, .025, .0125$. The covariance error is below 10^{-12} and the quadratic-generator discrepancy declines at the predicted rate. These tests include, rather than infer, the cross term. At a zero-adjustment interior state, the generator contribution of u^2 is .0025, not zero.

There is no nearest-neighbor rounding of the next preference state. The mean preference displacement is $\theta\Delta$ and bilinear continuation varies between nodes. A smooth test function linear in u therefore sees every nonzero feasible adjustment even when $|\theta|\Delta$ is less than half a grid spacing. Preference reachability is not inferred from the existence of a control parameter in metadata.

Positive bilinear interpolation on a smooth function has error of order $h_u^2 + h_X^2$ under appropriate derivative bounds. Dividing by Δ produces a generator contribution of order $(h_u^2 + h_X^2)/\Delta$. Thus an interior consistency limit requires a coupled choice such as $(h_u^2 + h_X^2)/\Delta \rightarrow 0$, together with vanishing action and time errors. Near the killed boundary, additional exit-time consistency and boundary regularity are needed. The four-branch linear-segment rule is not an exact Brownian-bridge crossing calculation. These qualifications explain why the displayed separate refinements are informative diagnostics but not a continuum error certificate.

B.3. Boundary cash flows and recorded exits

For a proposed increment d , the exit fraction is the minimum positive fraction at which $s + \lambda d$ reaches a domain face, capped at one. At a crossing the continuation value is $g(t + \lambda\Delta, s + \lambda d)$, the flow receives discount integral $[1 - \exp(-\rho\lambda\Delta)]/\rho$, and there is no continuation policy after settlement. A regression starts at $X = .51$, chooses $c = .8$, $\pi = 0$, and verifies that every branch reaches $X = .5$ with fraction strictly below one and then terminates. The program never changes wealth to $.5$ and permits another consumption date under the pre-exit contract.

The exit ledger distinguishes an exit event from a post-correction state. The adjustment budget is accumulated only until settlement. For a surviving branch interpolated to a boundary node, the next-date boundary settlement is part of the finite kernel. This interpolation effect is a numerical feature, not a physical reflection. The settlement fee is a model primitive; comparing fees 6, 8, and 10 is explicitly labeled a comparison of contracts.

APPENDIX C: RECURSIVE AGGREGATION, TEMPORAL SELVES, AND STRATEGIC INTERACTION

C.1. The recursive aggregator and its domain

For $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $bV > 0$, expand the aggregator as in (17). Its consumption derivative is

$$f_c(c, V) = \rho c^{a-1} (bV)^{1-a/b} > 0.$$

Its value derivative is

$$f_V(c, V) = \frac{\rho}{a} (b - a) c^a (bV)^{-a/b} - \frac{\rho b}{a}.$$

On compact ranges with consumption bounded away from zero and bV bounded away from zero, these derivatives are bounded. This is a local Lipschitz conclusion on the specified range, not a global sign or existence theorem for all values. The transform $V = e^W/b$ enforces $bV > 0$, but cannot by itself verify integrability, terminal compatibility, comparison, or transversality.

A policy-wise recursive evaluation solves (3); in the Markov setting its equation is $v_t + \mathcal{L}^\pi v + f(c^\pi, v) = 0$. When f is Lipschitz in v with constant L , standard comparison of a perturbed evaluation with a verified evaluation gives a finite-horizon error envelope of the form

$$e^{L(T-t)} b + \int_t^T e^{L(r-t)} \epsilon dr$$

under the relevant comparison and integrability hypotheses. A decreasing driver can give a sharper dissipative estimate, but monotonicity must be checked from f_V on the actual range. The time-separable discounted bound in Theorem 1 is not silently applied to an arbitrary Epstein–Zin driver.

C.2. Executed implicit recursive reference

The supplementary calculation fixes $\gamma = 5$, $\psi = 1.5$, and $\rho = .04$ on a deterministic saving problem with ten steps of length .05. At each wealth node $x \in [.5, 2]$, consumption is $c = mx$ with 80 fractions in $[.01, .6]$, and the next wealth is the stated saving transition in the code. For each action the scalar evaluation equation is

$$y = I_h V_{n+1}(x') + \Delta f(c, y), \quad by > 0.$$

The negative-value root is bracketed and solved numerically before maximizing over the action grid. The terminal target is x^b/b . Outside the wealth lattice, continuation uses the specified terminal homothetic tail as numerical boundary data at that date. It is not a solution on an unbounded wealth domain. At the central node the recorded value is $-.29658656$ and the selected consumption fraction is the lower bound .01. This active bound is reported rather than described as an interior stationary ratio.

The sign-domain and implicit-equation residual tests are actual evaluated quantities. Neural solution error is null because this experiment solves scalar implicit reference equations rather than training a recursive neural policy. The general recursive portfolio model remains part of the paper’s operator formulation and analytical derivations; its stationary ratios .3 and .0455 remain algebraic anchors until their full infinite-horizon verification is supplied.

C.3. Ordinary continuation and acting-self utility

For a fixed continuation policy, ordinary log-wealth utility in the two-consumption-date example has form $V_n(w) = A_n \log w + B_n$. Starting from $A_N = 1, B_N = 0$, the sophisticated self chooses

$$\alpha_n = \frac{1}{1 + \beta \delta A_{n+1}}, \quad c_n = \alpha_n w, \quad A_n = 1 + \delta A_{n+1}, \quad B_n = \log \alpha_n + \delta [A_{n+1} \log(1 - \alpha_n) + B_{n+1}].$$

Evaluation uses δ , while improvement uses $\beta \delta$. With $\delta = 1$ and two decisions, $A_1 = 2$ and $\alpha_0 = 1/(1 + 2\beta)$. Substituting β into the evaluation coefficient instead produces the different expression $1/(1 + \beta + \beta^2)$. At $\beta = 1$ the two formulas coincide, so this special case cannot diagnose the distinction.

A one-shot deviation is evaluated against the same frozen continuation policy whose equilibrium is being claimed. Replacing its continuation with a different grid’s optimized value changes the estimand. For a general finite sophisticated-self computation, all feasible actions can be enumerated at each date using the correctly evaluated future policy. For continuous actions a local search gives only a feasible deviation lower bound unless concavity, a global oracle, or an explicit upper-bound calculation closes the search gap.

C.4. Dynamic games and equilibrium selection

Let $J_i(\pi_i, \pi_{-i})$ denote player i ’s payoff with its own objective and transition law. A Nash certificate bounds

$$\sup_{\tilde{\pi}_i} J_i(\tilde{\pi}_i, \pi_{-i}) - J_i(\pi_i, \pi_{-i}),$$

not a joint-profit improvement. An actor graph that differentiates through rivals is not the unilateral derivative graph. In the static benchmark $q_i(1 - q_i - q_j)$, the derivative with a detached rival is $1 - 2q_i - q_j$. It vanishes at $q_i = q_j = 1/3$, whereas joint optimization selects $1/4$ and leaves a profitable deviation. The executable graph regression verifies both the own gradient and the absence of a rival parameter gradient.

The three-date capacity game has an exogenous Markov demand state and endogenous future capacity. At each state-date pair, enumerate the two players' current feasible output–investment actions using continuation values from the already selected future equilibria. A pure equilibrium is an action pair that is a best response for each player against the other current action and the fixed future policy profile. Every enumerated state has at least one pure equilibrium in this particular game. Lexicographic tie selection is recorded, and the full profile includes asymmetric choices. In a finite-horizon game, this backward best-response construction establishes subgame perfection on the finite state space; no uniqueness follows from it.

For large-population or competitive-limit analysis, the demand and capacity scaling, investment technology, and equilibrium concept must be fixed before taking a limit. Symmetric initialization is neither a proof of equilibrium nor an equilibrium-selection theorem. Those questions are retained as model extensions rather than filled with the historical illustrative N -series. No claim about a trained high-dimensional game or a solved competitive limit is attached to the current exact finite game.

APPENDIX D: TRACE ESTIMANDS AND COMPUTATIONAL COST

For standard Gaussian z and symmetric A , diagonalization and the independent squared-normal moments give $\mathbb{E}z'Az = \text{tr } A$ and $\text{Var}(z'Az) = 2\|A\|_F^2$. Independence across K probes yields variance $2\|A\|_F^2/K$ for their mean. Expanding the squared residual then proves (20). For Rademacher probes the diagonal contribution to variance vanishes, so the formula must be changed rather than reused uncritically.

For $K \geq 2$, define the cross-product correction

$$U_K = a^2 - a\bar{q}_K + \frac{1}{4K(K-1)} \sum_{i \neq j} q_i q_j. \quad (22)$$

Independence of distinct probes gives $\mathbb{E}U_K = (a - \text{tr } A/2)^2$. The statistic is unbiased but can be negative in a finite batch and may have a different variance and optimization behavior. It is therefore an additional estimand, not a blanket replacement justified by unbiasedness alone. The audit reports its empirical mean and standard error separately. When A depends on network parameters, the uncorrected variance term itself varies with those parameters; a squared stochastic trace can consequently change the expected minimizer.

Let C_{jet} denote the actual cost of a directional second derivative and its parameter gradient for the chosen network. If Σ has m columns, exact contraction can be organized as m directional second derivatives, with loading and batching costs. A K -probe contraction costs roughly $K C_{\text{jet}}$ plus those costs; it is not an unconditional $O(d)$ solver. Total work additionally multiplies by collocation samples, training iterations, and any restarts. Peak memory, warm-up, and accelerator synchronization must be measured for a hardware comparison. The current CPU timers do not measure peak memory, so that metric remains null.

D.0.0.1. The coupled dynamic regression. For the finite-horizon quadratic-cost economy, a policy $a = Wx$ gives a quadratic value $x'Px + c$. Given a frozen next evaluator, the actor minimizes the expected quadratic stage-plus-continuation cost. Gaussian shocks contribute

$\text{tr}(PSS')$; the value recursion includes this constant. A matrix-valued linear neural actor is trained against the frozen evaluator by L-BFGS, while an independent Riccati recursion computes the benchmark. Both use identical A, Q, R, S , horizon and terminal cost. Correlation is a state-transition covariance here, not a metadata label on a static objective.

The known quadratic family makes this a favorable regression. At dimensions 4, 8, and 16, Riccati times are approximately .00060, .00040, .00043 seconds and actor times are 1.319, .0323, .0384 seconds. The first timing includes startup effects. These numbers neither establish a dimension scaling law nor compare against adaptive sparse grids. The original static coupled quadratic program is retained in the R3 archive as a small linear-algebra check, separate from this dynamic experiment.

APPENDIX E: APPROXIMATION, COUNTEREXAMPLES, AND OPTIMIZATION LIMITS

E.1. *A composite objective can bias evaluation*

Consider a one-state terminal evaluation problem whose true value is zero and whose critic is a scalar v . The exact evaluation residual is v . A joint surrogate $v^2 - \lambda v$ has minimizer $v = \lambda/2$, not the evaluated value, for every $\lambda \neq 0$. The example isolates the logical issue: optimizing performance and evaluation over the same value parameter need not preserve the evaluation equation. Adding a positive residual weight changes the bias but does not prove exact evaluation. The separated actor has no gradient into that value parameter, so it does not create this particular bias.

Similarly, zero-centering a residual is not a stability theorem. The derivative of $R^2/2$ is $R\nabla R$ and its Jacobian, conditioning, sampling, and step sizes govern a finite optimizer. Value dependence in a recursive driver remains inside this derivative. A zero target does not make the objective independent of the value level.

E.2. *A small squared residual does not select a viscosity solution*

On $(-1, 1)$ with boundary values zero, consider $|v'| = 1$. The nonnegative distance function $1 - |x|$ is the viscosity solution. The function $|x| - 1$ has the same boundary values and satisfies $|v'| = 1$ almost everywhere, but it fails the viscosity supersolution test at zero: the constant test function -1 touches it from below and has $|\phi'| - 1 = -1 < 0$.

Smooth approximations $v_\epsilon(x) = \sqrt{x^2 + \epsilon^2} - \sqrt{1 + \epsilon^2}$ have exactly the boundary data and converge uniformly to $|x| - 1$. Their squared residual $\int_{-1}^1 (|v'_\epsilon| - 1)^2 dx$ tends to zero by dominated convergence, although the limit is not the viscosity solution. Thus boundary agreement, smooth networks, and an L^2 residual do not establish viscosity selection. Monotone, stable, consistent schemes with appropriate comparison can provide a different route (Barles and Souganidis, 1991); positivity of an interior interpolation alone does not verify all of those hypotheses at a killed boundary.

Smooth nonpolynomial neural networks can approximate a sufficiently smooth function and derivatives on compact subsets under the relevant approximation hypotheses. This does not create derivatives at a kink, ensure global boundary regularity, close a finite policy class under a maximizing selector, or prove that stochastic optimization finds the approximant. The retained derivative-approximation motivation is used only in that conditional sense.

E.3. *Two-timescale statements and suboptimal attractors*

For projected stochastic approximation, a conventional ODE analysis requires bounded iterates, suitable continuity and stability, martingale-difference noise with bounded conditional

second moments, step sizes satisfying infinite sums, finite squared sums, and a vanishing slow-to-fast step-size ratio, plus a globally attracting critic equilibrium for each frozen actor. Under the appropriate additional technical conditions, the actor limit set is governed by the corresponding limiting ODE and its internally chain-transitive sets.

This is not a global optimality statement. For $H(a) = -(a^2 - 1)^2 + .2a$, the negative local maximizer is an asymptotically stable isolated stationary point with strictly positive global improvement gap. The audit solves the cubic stationarity equation and evaluates the gap .3998745872. Exact critic evaluation and zero noise do not remove that attractor. To conclude optimality one must establish, for the actual actor dynamics, that every relevant limiting set has zero feasible improvement gap, or use an applicable global concavity or gradient-domination argument. The finite runs use Adam and L-BFGS and are assessed by their measured errors; they are not examples of a diminishing-step asymptotic theorem.

APPENDIX F: IMPLEMENTATION, EVIDENCE CLASSES, AND REPRODUCIBILITY

F.1. *Executed parameterizations and data flow*

The continuous Merton critic has one learned log-amplitude and its actor has two learned logits. It uses float64 automatic differentiation through the value with respect to wealth, including the Hessian; critic and actor optimizers are separate. Its independently evaluated policy has positive transversality decay D . Analytical controls are used only after training to measure error.

The NDU run is different: time-specific nonlinear networks represent a finite Bellman operator. Its actor maps two normalized states to 125 logits through two 48-unit tanh layers. Its critic maps six features, including log distances to boundary faces, through two 48-unit tanh layers to a scalar adjustment to terminal utility. Next-date critics are frozen, the actor target uses only their continuation predictions, and the current critic is fitted to the enacted actor's policy-evaluation target. The independent NumPy optimizer and evaluator are not imported as training labels.

Training residuals are not out-of-sample certificates because all interior finite states are collocation points. The independent evaluator changes the computational path, not the economic grid. The source contains separate NumPy and PyTorch implementations of the transition and interpolation, whose equality is tested on deterministic data. Cross-grid reference calculations are discretization diagnostics; they have not been relabeled a held-out neural accuracy result.

F.2. *Interpreting metrics*

A value error against the finite optimal reference, a critic error against the evaluated policy, a coarse-fine value difference, and a unilateral gain are different estimands. Policy component differences are retained for consumption, adjustment, and portfolio share. Boundary-node actor outputs are not enacted after settlement; full-grid maxima and interior summaries must be distinguished when interpreting them. A finite exhaustive best-response maximum is a finite-model certificate. A feasible search over only some continuous deviations is merely a lower bound on exploitability.

Execution status and tolerance status are separate fields. A program that completes can miss its target; a timed-out audit has no final solution certificate. A null numerical error has an explicit reason, such as an unmeasured continuous-diffusion error or a recursive reference that did not train a neural policy. A numerical zero is used only for a quantity actually calculated, such as the finite game deviation maximum or a tested monotonicity violation.

The initial failed neural source and its weights, metrics, and raw arrays are retained in the local review bundle. The successful three-seed sources and weights are separate. The attempted twelve-date runs are marked incomplete. Wall-clock times are metadata and are excluded from

scientific payload hashes. Source hashes identify the exact executed script; the reachable review commit identifies the historical base, not a fictitious commit containing the subsequently created scripts. The final manifest maps every delivered source to its actual file hash and repository path.

F.3. *Retained material and revision lineage*

The new authoritative entry is `ECTA_R4.tex`. The original `ECTA.tex`, the R3-authoritative `ECTA_R2.tex`, its standalone supplement, embedded arrays, earlier reports, and R3 numerical artifacts are left unchanged. Core model formulations, constrained policy derivations, recursive aggregation, temporal-self and game extensions, trace analysis, and theoretical qualifications appear in the current text and these appendices. Historical unexecuted plots are retained as archival objects, not presented again as learned current outcomes. The retention map identifies their source locations and the corresponding corrected discussion. This preserves the scientific record without repeating incompatible equations as simultaneous current specifications.

PART IV: DELIVERED R5 NUMERICAL RECORD

The tables below reproduce all six delivered R5 TeX tables without changing their entries. They refer to the source and protocols reviewed on September 21, 2026. The R8 main paper gives the current interpretation: finite candidate certificates are not continuous certificates; safeguarded rules are distinct from raw rules; the in-house PINN–PI comparison is not the new author-code comparison; and these historical extension panels have only two seeds. The execution data, raw arrays, checkpoints, and failed or incomplete attempts remain in the corresponding revision directories.

TABLE I
RETAINED R5 NDU PANEL

Configuration	n	Raw	Safeguarded	Max. bound	Seconds
baseline	10	10	10	0.0496	18.19
longer horizon	3	0	3	0.0064	34.64
width 32	2	1	2	0.0471	16.86
width 64	2	2	2	0.0461	20.03
lr 0.002	2	0	2	0.0050	18.52
lr 0.008	2	2	2	0.0339	17.95
actor steps 300	2	0	2	0.0065	13.66
critic steps 400	2	1	2	0.0440	15.70

In this table, n denotes the number of runs, not the number of time steps. Pass counts use the historical finite-candidate target. Seconds are the historical recorded training measure; this is not the new full safeguard re-audit timing.

TABLE II
RETAINED R5 QUADRATIC-CONTROL COMPARISON

d	Method	Mean loss	Sampling upper	Train (s)	Audit (s)	MiB
8	NBO	0.0010	0.0018	14.93	3.45	377.5
8	PINN–PI	-0.0001	0.0007	12.59	6.61	376.9
16	NBO	0.0037	0.0055	14.29	3.84	378.6
16	PINN–PI	0.0013	0.0032	11.67	5.69	377.7
32	NBO	0.0102	0.0145	16.29	3.72	381.3
32	PINN–PI	0.0075	0.0117	13.34	5.64	382.1

This is the two-seed Cole–Hopf/QMC benchmark with an analytic greedy control and an in-house PINN–PI comparator. The small negative estimated loss reflects reference or simulation uncertainty, not performance exceeding a true optimum. NBO has higher mean loss in each row pair. This result is retained despite the favorable R8 nonconvex-control comparison.

TABLE III
RETAINED R5 MANUFACTURED STOPPED-DIFFUSION TEST

N	Endpoint value	Bridge value	Action sup error	Policy-loss bound
16	0.0392	0.0581	0.3783	0.1691
32	0.0616	0.0308	0.4653	0.0737
64	0.0623	0.0159	0.5033	0.0366
128	0.0538	0.0080	0.5346	0.0219
256	0.0436	0.0040	0.5502	0.0106

The historical one-dimensional solver has an analytically manufactured verification pair and no trained actor. It is distinct from the new learned NDU-geometry test, which also explicitly modifies the running payoff.

TABLE IV
RETAINED R5 COUPLED NDU REFINEMENTS

N	State grid	n_a	h^2/Δ	V_0	B_0	Exit
12	13×16	3	0.2133	-1.624355	0.012266	0.1543
24	25×31	3	0.1067	-1.510132	0.012248	0.1558
48	49×61	5	0.0533	-1.391646	0.011969	0.1024
96	97×121	5	0.0267	-1.353683	0.012234	0.0987

Multiple discretization factors change in this historical panel. The table supplies numerical differences, not validated continuous error bounds or a separated convergence order.

TABLE V
RETAINED R5 RECURSIVE-UTILITY POLICIES

Seed	Raw log loss	Final bound	Final log loss	m_0	p_0
51	0.0425	0.0000	0.0000	2.506	0.955
52	0.0502	0.0000	0.0000	2.442	1.093

Both raw neural policies miss the historical .01 log-value target. The final candidate-set bounds use repeated 961-action safeguards; the raw failures are not relabeled as neural successes.

TABLE VI
RETAINED R5 NEURAL GAME

Seed	Player 1, $t = 0$	Player 2, $t = 0$	All subgames	Seconds
61	0.00245	0.00673	0.00673	13.87
62	0.00312	0.00305	0.00368	14.04

This is a small finite-state, three-date game with exact continuous best-response auditing for the declared finite operator, not a continuous-state equilibrium scalability result.

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